Exercise 12.2.4 Rederive the equivalent of Eq. (12.2.23) keeping terms of order $\epsilon_x\epsilon_z^2$ (You may assume that $\epsilon_y = 0$). Use this information to rewrite Eq. (12.2.24) to order $\epsilon_x\epsilon_z^2$. By equating coefficients of this term deduce the constraint

$$-2L_zP_xL_z + P_xL_z^2 + L^2P_x = \hbar^2P_x$$

This seems to conflict with the statement (1) made above, but not really in view of the identity

$$-2ABA + BA^2 + A^2B = [A,[A,B]]$$

Using this identity, verify that the new constraint coming from the $\epsilon_x\epsilon_z^2$ term is satisfied given the commutation relations between P_x, P_y, L_z.

First we recognize that since we work on the order $\epsilon_x\epsilon_z^2$, it is necessary that any small angle expansion goes to a similar order. That is that for small θ, $\cos(\theta) \approx 1-\theta^2/2$ and $\sin(\theta) \approx \theta$. This means that a coordinates in x-y are transformed by a rotation ϕ along $\vec{z}$ by

$$x' = (1-\phi^2/2)x - \phi y$$

and

$$y' = (1-\phi^2/2)y + \phi x$$

Also operators are exponentiated according to and $e^A \approx 1 + A + \frac{1}{2}A^2$.

Now, Let us consider the transformation $T^* \equiv U[R(-\epsilon_z\vec{k})]T(-\epsilon_x\vec{x})U[R(\epsilon_z\vec{k})]T(\epsilon_x\vec{x})$

$$\begin{bmatrix}x\\y\end{bmatrix} \xrightarrow{\epsilon_x} \begin{bmatrix}x+\epsilon_x\\y\end{bmatrix} \xrightarrow{R(\epsilon_z\vec{k})} \begin{bmatrix}(1-\epsilon_z^2/2)(x+\epsilon_x)-\epsilon_z y\\(1-\epsilon_z^2/2)y+\epsilon_z(x+\epsilon_x)\end{bmatrix}$$
$$\xrightarrow{-\epsilon_x} \begin{bmatrix}x-\epsilon_z y-\epsilon_z^2x/2-\epsilon_x\epsilon_z^2/2\\ y+\epsilon_z x-\epsilon_z^2y/2+\epsilon_x\epsilon_z\end{bmatrix}$$
$$\xrightarrow{R(-\epsilon_z\vec{k})} \begin{bmatrix}(1-\epsilon_z^2/2)(x-\epsilon_z y-\epsilon_z^2x/2-\epsilon_x\epsilon_z^2/2)+\epsilon_z(y+\epsilon_z x-\epsilon_z^2y/2+\epsilon_x\epsilon_z)\\(1-\epsilon_z^2/2)(y+\epsilon_z x-\epsilon_z^2y/2+\epsilon_x\epsilon_z)-\epsilon_z(x-\epsilon_z y-\epsilon_z^2x/2-\epsilon_x\epsilon_z^2/2)\end{bmatrix}$$
$$= \begin{bmatrix}x+\epsilon_x\epsilon_z^2/2\\ y+\epsilon_x\epsilon_z\end{bmatrix}$$

Now in a basis independent representation, we expand the transformation. We will denote $A \equiv \frac{i}{\hbar}\epsilon_z L_z$ and $B \equiv \frac{i}{\hbar}\epsilon_x P_x$. Note that $[A,B]$ can be computed using previously shown relations (12.2.17) and is equal to $-\frac{\epsilon_x\epsilon_z i}{\hbar}P_y$.

We rexpress the compounded transformation T^*. Note that because of our order restriction, we throw out any terms that would produce ϵ_x^2, ϵ_z^3, or any higher orders.

$$\begin{aligned}
T^* &= (\mathbb{1} + A + \frac{1}{2}A^2)(\mathbb{1}+B)(\mathbb{1}-A+\frac{1}{2}A^2)(\mathbb{1}-B)\\
&= (\mathbb{1}+B+A+AB+\frac{1}{2}A^2+A^2B)(\mathbb{1}-B-A+AB+\frac{1}{2}A^2-\frac{1}{2}A^2B)\\
&= \mathbb{1} + [A,B] + \frac{1}{2}BA^2 + \frac{1}{2}A^2B - ABA\\
&= \mathbb{1} - \frac{i}{\hbar}\epsilon_x\epsilon_z P_y +
\end{aligned}$$

Now we can bring about trying to relate the two approaches. Notice that the geometric representation reveals that the 4-fold transformation ammounts to a single transformation $T^* = \mathbb{1} - \frac{i}{h}((\epsilon_x\epsilon_z^2/2)P_x - (\epsilon_x\epsilon_z)P_y)$.

In this way, we can notice how the P_y term is explicitly clear in both approaches. This leaves how the remaining $\frac{1}{2}BA^2 + \frac{1}{2}A^2B - ABA$ term is equivalent to the P_x coefficient.

We assert that

$$\frac{1}{2}BA^2 + \frac{1}{2}A^2B - ABA = -\frac{i}{\hbar}(\epsilon_x\epsilon_z^2/2)P_x$$

We can multiple both sides by $2i\hbar^3/\epsilon_x\epsilon_z^2$ and re-input A and B to get

$$P_xL_z^2 + L_z^2P_x - 2L_zP_xL_z = \hbar^2P_x$$

In another point of view, we can identify the identity stated in the premise to reveal that the LHS above is also

$$\frac{1}{2}[A,[A,B]] = -\frac{1}{2\hbar^2}\epsilon_x\epsilon_z^2[L_z,P_y] = \frac{i}{\hbar}(\epsilon_x\epsilon_z^2/2)Px$$

Exercise 12.3.1 We wish to enforce the hermiticity constraint on the angular momentum operator L_z. This is represented by imposing $\langle\psi_1|\, L_z\, |\psi_2\rangle = \langle\psi_2|\, L_z\, |\psi_1\rangle^*$ (Eq. 12.3.5 to 12.3.6).

In the coordinate basis, the condition is,

$$\int_0^\infty\int_0^{2\pi}\psi_1^*\Big(-i\hbar\frac{\partial}{\partial\phi}\Big)\psi_2\rho d\rho d\phi = \Big[\int_0^\infty\int_0^{2\pi}\psi_2^*\Big(-i\hbar\frac{\partial}{\partial\phi}\Big)\psi_1\rho d\rho d\phi\Big]^*$$

On the left hand side we integrate by parts

$$\begin{aligned}
&= \int_0^\infty\Big[-i\hbar\psi_1^*\psi_2\Big|_0^{2\pi} + i\hbar\int_0^{2\pi}\psi_2\frac{\partial\psi_1^*}{\partial\phi}d\phi\Big]\rho d\rho \\
&= \Big[\int_0^\infty\Big[i\hbar\psi_1\psi_2^*\Big|_0^{2\pi} - i\hbar\int_0^{2\pi}\psi_2^*\frac{\partial\psi_1}{\partial\phi}d\phi\Big]\rho d\rho\Big]^*
\end{aligned}$$

Notice that the LHS is equivalent to the right hand side on the condition that $i\hbar\psi_1\psi_2^*\big|_0^{2\pi} = 0$. This condition is equivalent to requiring that each ψ obeys

$$\psi(\rho,0) = \psi(\rho,2\pi)$$

which is the desired conclusion.

Exercise 12.3.2 Let us try to deduce the restriction on l_z from another angle. Consider the superposition of two allowed l_z eigenstates:

$$\psi(\rho,\phi) = A(\rho)e^{i\phi l_z/\hbar} + B(\rho)e^{i\phi l_z'/\hbar}$$

.

We will demand that upon a 2π rotation, we get the same physical state (equivalent up to a global phase Φ).

$$\psi(\rho, 2\pi) = \psi(\rho, 0)e^{i\Phi}$$
$$A(\rho)e^{i2\pi l_z/\hbar} + B(\rho)e^{i2\pi l'_z/\hbar} = (A(\rho) + B(\rho))e^{i\Phi}$$

Matching coefficients we enforce that

$$e^{i2\pi l_z/\hbar} = e^{i\Phi} = e^{i2\pi l'_z/\hbar}$$
$$e^{i2\pi(l_z - l'_z)/\hbar} = 1$$

From this condition, we see that $l_z - l'_z = m\hbar$ where m is an integer. Then from symmetry, as the system could be rotated to allow a angular momentum in the opposite direction, then these integer values should be allowed to be negative as well. In order that m is an integer, we insist that l_z and l'_z are integer multiples of $\hbar$ or half integers of $\hbar$. i.e. $l_z \in \{..., 3\hbar/2, \hbar/2, -\hbar/2, -3\hbar/2...\}$ or $l_z \in \{...2\hbar, \hbar, 0, -\hbar, -2\hbar...\}$.

Exercise 12.3.3 A particle is described by a wave function

$$\psi(\rho, \phi) = Ae^{-\rho^2/2\Delta^2}\cos^2\phi$$

We will calculate the probabilities of each of the available eigenvalues of L_z. Immediately, we can see that the wave function is separable into, first, its radial part and then second, the azimuthal part.

Let us first note that the $\cos^2\phi$ term can be expressed in terms of the eigenfunction of L_z, $\Phi_m(\phi) = (2\pi)^{-1/2}e^{im\phi}$, by using the euler identity $\cos\phi = (e^{i\phi} + e^{-i\phi})/2$.

$$\begin{aligned}\cos^2\phi &= \frac{1}{4}\left(2 + e^{2i\phi} + e^{-2i\phi}\right) \\ &= \sqrt{2\pi}\left(\frac{1}{2}\Phi_0(\phi) + \frac{1}{4}\Phi_2(\phi) + \frac{1}{4}\Phi_{-2}(\phi)\right)\end{aligned}$$

This tells us that the azimuthal part can be expressed as three different eigenfunction of eigenvalues $l_z = 2\hbar, 0, -2\hbar$.

Additionally, the eigenfunctions of L_z, in the coordinate basis, satisfy

$$-i\hbar\frac{\partial\psi_{l_z}(\rho, \phi)}{\partial\phi} = l_z\psi_{l_z}(\rho, \phi)$$

Thus, the corresponding eigenfunctions of L_z satisfying this condition are $\psi_0 = R(\rho)$, $\psi_{2\hbar} = R(\rho)e^{2i\phi}$, and $\psi_{-2\hbar} = R(\rho)e^{-2i\phi}$

Finally, we can write the probability, $P(l_z = l_i) = |\langle l_i|\psi\rangle|^2$. We will compute the scalar $\langle l_z|\psi\rangle$ by inserting the coordinate identity. (this is slightly inaccurate, the notation would imply a particular state for each l_z which is not quite accurate, which will be seen later)

$$\langle l_i|\psi\rangle = \int_0^{2\pi}\int_0^{\infty} \rho d\rho d\phi \, \langle l_i|\rho,\phi\rangle \, \langle \rho,\phi|\psi\rangle = \int_0^{2\pi}\int_0^{\infty} \rho d\rho d\phi \psi_{l_z}^* A e^{-\rho^2/2\Delta^2} \cos^2\phi$$
$$= \int_0^{2\pi}\int_0^{\infty} \rho d\rho d\phi \psi_{l_z}^* A e^{-\rho^2/2\Delta^2} \left(\sqrt{2\pi}\left(\frac{1}{2}\Phi_0(\phi) + \frac{1}{4}\Phi_2(\phi) + \frac{1}{4}\Phi_{-2}(\phi)\right)\right)$$

The eigenvalue $l_z = m\hbar$ is degenerate because the corresponding eigenfunctions can have any radial dependence $R(\rho)$. However, since the same radial factor appears in every term of our wavefunction, it contributes equally to the norm of each angular momentum component and therefore cancels when calculating the relative probabilities. Thus we can proceed to just sum the squared coefficients of these orthonormal eigenfunctions, and using it to normalize our values. (what is not shown is taking $\langle l_i|\psi\rangle$, and multiplying its complex conjugate, and using the orthonormality of Φ in order to leave only the squared coefficients of each Φ.

Hence,

$$P(l_z = 2\hbar) = \frac{(1/4)^2}{(\frac{1}{4}+\frac{1}{16}+\frac{1}{16})} = 1/6$$
$$P(l_z = 0) = \frac{(1/2)^2}{(\frac{1}{4}+\frac{1}{16}+\frac{1}{16})} = 2/3$$
$$P(l_z = -2\hbar) = \frac{(1/4)^2}{(\frac{1}{4}+\frac{1}{16}+\frac{1}{16})} = 1/6$$

Exercise 12.3.4 A similar case can be done with a particle described by wave function

$$\psi(\rho,\phi) = Ae^{-\rho^2/2\Delta^2}\left(\frac{\rho}{\Delta}\cos\phi + \sin\phi\right)$$

While this wavefunction is not immediately separable, it is the sum of separable functions. Rearranging it we get

$$\psi = \frac{\rho A e^{-\rho^2/2\Delta^2}}{2\Delta}\left(e^{i\phi}+e^{-i\phi}\right) + \frac{Ae^{-\rho^2/2\Delta^2}}{2i}\left(e^{i\phi}-e^{-i\phi}\right)$$
$$\psi = \frac{Ae^{-\rho^2/2\Delta^2}}{2}\left[\left(\frac{\rho}{\Delta}-i\right)e^{i\phi} + \left(\frac{\rho}{\Delta}+i\right)e^{-i\phi}\right]$$

Like above, the $e^{\pm i\phi}$ references the angular eigenfunctions of L_z and hints that the only available eigenvalues take on $l_z = \pm\hbar$.

Like above, the factors outside of [...] do not affect the relative probabilities. We can continue to calculate the probabilities of each of the eigenvalues by doing the same process of squaring the coefficients and normalizing it (and taking care of any conjugations that occur when squaring these complex numbers).

$$P(l_z = \hbar) = \frac{|\frac{\rho}{\Delta}-i|^2}{|\frac{\rho}{\Delta}-i|^2 + |\frac{\rho}{\Delta}+i|^2} = P(l_z = -\hbar) = \frac{\frac{\rho^2}{\Delta^2}+1}{2[\frac{\rho^2}{\Delta^2}+1]} = \frac{1}{2}$$

Exercise 12.3.5 Note that the angular momentum seems to generate a repulsive potential in the radial equation, that determines $R_{Em}(\rho)$ and the allowed values for E,

$$\left[\frac{-h^2}{2\mu}\left(\frac{d^2}{d\rho^2}+\frac{1}{\rho}\frac{d}{d\rho}-\frac{m^2}{\rho^2}\right)+V(\rho)\right]R_{Em}(\rho)=ER_{Em}(\rho)$$

Calculate its gradient and identify it as the centrifugal force.

If we compare this Hamiltonian to the generic Hamiltonian for rotationally invariant problems

$$\left[\frac{-h^2}{2\mu}\left(\frac{\partial^2}{\partial\rho^2}+\frac{1}{\rho}\frac{\partial}{\partial\rho}+\frac{1}{\rho^2}\frac{\partial^2}{\partial\phi^2}\right)+V(\rho)\right]$$

we notice that the constant $-m^2/\rho^2$ would closely resemble the angular kinetic energy $\frac{1}{\rho^2}\frac{\partial^2}{\partial\phi^2}$.

Just like in classical mechanics, we can treat this angular kinetic energy as part of an "effective" potential by slightly rearranging the equation and declaring an effective potential $V_{eff}(\rho)$

$$V_{eff}(\rho)=\frac{\hbar^2m^2}{2\mu\rho^2}+V(\rho)$$

Then from classical mechanics, we can find the force a potential produces.

$$\begin{aligned}F&=-\nabla V_{eff}\\&=-\left(\vec{\rho}\frac{\partial}{\partial\rho}+\vec{\phi}\frac{1}{\rho}\frac{\partial}{\partial\phi}\right)\left(\frac{\hbar^2m^2}{2\mu\rho^2}+V(\rho)\right)\\&=\frac{\hbar^2m^2}{\mu\rho^3}\vec{\rho}-\frac{dV(\rho)}{d\rho}\vec{\rho}\end{aligned}$$

In this way, we can see that the presence of angular kinetic energy and hence, angular momentum produces a centrifugal force in positive ρ direction, while the potential itself produces a force in the negative ρ direction.

Exercise 12.3.6 Consider a particle of mass μ constrained to move on a circle of radius a. Show that $H=L_z^2/2\mu a^2$. Solve the eigenvalue problem of H and interpret the degeneracy

---

Consider the radial equation from exercise 12.3.5 and the H embedded within it. A mass constrained to radius a will have a radial equation constant in ρ, thus any differential with respect to ρ on $R_{em}(a)$ will yield zero. The remaining Hamiltonian yields

$$\frac{\hbar^2}{2\mu}\frac{m^2}{a^2}+V(a)$$

after substituting ρ for a. Additionally, $V(a)$ will simply increase all energy levels by some constant $V(a)$, thus we can arbitrarily set it to zero. Remembering that $L_z=m\hbar$, we arrive at $H=L_z^2/2\mu a^2$

To solve the H eigenvalue problem we plug this H into the eigenvalue equation.

$$H\Phi_m(\phi) = \frac{m^2\hbar^2\Phi_m(\phi)}{2\mu a^2} = E_m\Phi_m(\phi)$$

Here we can see that E_m, the angular momentum energy is two fold degenerate since $+m$ and $-m$ yield the same values. Physically this corresponds to a particle moving clockwise or counter clockwise in a circle at radius a, and physical intuition would agree that they have the same energy.

The same analysis can be done by using the full wave function eigenvalue problem

$$\begin{aligned} H\psi_E &= E\psi_E \\ -\frac{\hbar^2}{2\mu a^2}\frac{\partial^2}{\partial\rho^2}\psi &= E\psi_E \end{aligned}$$

This differential equation has solutions of the form

$$\psi_E = Ae^{-i\hbar\phi/\sqrt{2\mu a^2 E}} + Be^{i\hbar\phi/\sqrt{2\mu a^2 E}}$$

.

On physical grounds we can restrict the wavefunction by asserting that $\psi(\phi) = \psi(\phi + 2\pi)$. After applying this restriction we can evaluate A and B and find the same solution as above.

Exercise 12.3.7 (The Isotropic Oscillator). Consider the Hamiltonian

$$H = \frac{P_x^2 + P_y^2}{2\mu} + \frac{1}{2}\mu\omega^2(X^2 + Y^2)$$

Part 1 Convince yourself that $[H, L_z] = 0$ and reduce the eigenvalue problem of H to the radial differential equation for $R_{Em}(\rho)$.

First, we notice that

$$\begin{aligned} [P_x^2, L_z] + [P_y^2, L_z] &= P_x[P_x, L_z] + [P_x, L_z]P_x + P_y[P_y, L_z] + [P_y, L_z]P_y \\ &= -i\hbar(P_xP_y + P_yP_x) + i\hbar(P_yP_x + P_xP_y) = 0 \end{aligned}$$

Similarly,

$$\begin{aligned} [X^2, L_z] + [Y^2, L_z] &= X[X, L_z] + [X, L_z]X + Y[Y, L_z] + [Y, L_z]Y \\ &= -i\hbar(XY + YX) + i\hbar(YX, XY) = 0 \end{aligned}$$

Then,

$$[H, L_z] = \frac{[P_x^2, L_z] + [P_y^2, L_z]}{2\mu} + \frac{1}{2}\mu\omega^2([X^2, L_z] + [Y^2, L_z]) = 0$$

Now, we take notice that the second term, in polar coordinates, is the potential $V(\rho) = \frac{1}{2}\mu\omega^2\rho^2$, then radial differential equation now takes the form,

$$\left[\frac{-\hbar^2}{2\mu}\left(\frac{d^2}{d\rho^2} + \frac{1}{\rho}\frac{d}{d\rho} - \frac{m^2}{\rho^2}\right) + \frac{1}{2}\mu\omega^2\rho^2\right]R_{Em}(\rho) = ER_{Em}(\rho)$$

Part 2 Examine the radial equation as $\rho \to 0$ and show that $R_{Em}(\rho) \xrightarrow{\rho \to 0} \rho^{|m|}$

Here, we want to asymptotically anaylse the behavior of this differential equation and make a guess for its solution. Luckily, the problem gives us a hint with $\rho^{|m|}$. Let's assume that in this regime, $R_{Em}(\rho) \to \rho^s$, where s is yet to be determined. Then $R'_{Em} = s\rho^{s-1}$ and $R''_{Em} = (s^2 - s)\rho^{s-2}$

Rewriting the radial equation, letting $\Gamma = \frac{-\hbar^2}{2\mu}$ for convenience, we get

$$\Gamma R''_{Em} + \frac{\Gamma}{\rho} R'_{Em} - \Gamma \frac{m^2}{\rho^2} R_{Em} + \frac{1}{2}\mu\omega^2\rho^2 R_{Em} = E R_{Em}$$

Or with our guess,

$$\Gamma\big[(s^2 - s) + s - m^2\big]\rho^{s-2} + \frac{1}{2}\mu\omega^2\rho^{s+2} = E\rho^s$$

As $\rho \to 0$, the ρ^{s-2} term becomes dominant and we can treat the remaining terms as zero. Solving for s yields,

$$\begin{aligned} \Gamma(s^2 - s + s - m^2)\rho^{s-2} &= 0 \\ s^2 - m^2 &= 0 \\ s &= \pm m \end{aligned}$$

Combining these solutions mean that $R_{Em} \xrightarrow{\rho \to 0} \rho^{|m|}$

Part 3 Show likewise that up to powers of ρ that $R_{Em}(\rho) \xrightarrow{\rho \to \infty} e^{-\mu\omega\rho^2/2\hbar}$

Similarly, we will assume a solution of $R_{Em} \to e^{-A\rho^2}$, and solve for A later. Then $R'_{Em} = -2A\rho R_{Em}$ and $R''_{Em} = (4A^2\rho^2 - 2A)R_{Em}$.

Plugging these into our radial equation we get

$$\left[\Gamma\big(4A^2\rho^2 - 2A - 2A - \frac{m^2}{\rho^2}\big) + \frac{1}{2}\mu\omega^2\rho^2\right] e^{-A\rho^2} = E e^{-A\rho^2}$$

We cancel out the R_{Em} term and take notice that ρ^2 terms become dominant as $\rho \to \infty$ over the lower power terms such that

$$(\Gamma 4A^2 + \frac{1}{2}\mu\omega^2)\rho^2 = 0$$

Thus solving for A yields $A = \pm\frac{\mu\omega}{2\hbar}$. Then we discard the negative solution as that yields a R_{Em} that does not decay as $\rho \to \infty$. Thus $R_{Em}(\rho) \xrightarrow{\rho \to \infty} e^{-\mu\omega\rho^2/2\hbar}$

Combining parts (2) and (3), We can now assume that $R_{Em}(\rho) = \rho^{|m|} e^{-\mu\omega\rho^2/2\hbar} U_{Em}(\rho)$

Part 4 Switch to the dimensionless variables $\epsilon = E/\hbar\omega$, $y = \sqrt{\mu\omega/\hbar}\rho$

Then our assumed radial function is $R_{Em} = \big(y\sqrt{\frac{\hbar}{\mu\omega}}\big)^{|m|} e^{-y^2/2} U^*_{Em}(y)$

Part 5 Convert the equation for R into an equation for U. I suggest proceeding in two stages: $R = y^{|m|}f$, $f = e^{-y^2/2}U$

Let's convert the radial equation to our new coordinates with the change of differentials $\frac{d}{d\rho} = \frac{d}{dy}\frac{dy}{d\rho} = \sqrt{\frac{\mu\omega}{\hbar}}\frac{d}{dy}$. Then,

$$\left[\frac{-\hbar^2}{2\mu}\left(\frac{\mu\omega}{\hbar}\frac{d^2}{dy^2} + \frac{\mu\omega}{\hbar}\frac{1}{y}\frac{d}{dy} - \frac{\mu\omega}{\hbar}\frac{m^2}{y^2}\right) + \frac{1}{2}\mu\omega^2 y^2\frac{\hbar}{\mu\omega}\right]R_{Em} = ER_{Em}$$

Lets now make some cancellations and substitutions, plug in $R_{Em} = y^{|m|}f$, then proceed to calculate the derivatives. Note that $R'_{Em} = |m|y^{|m|-1}f + y^{|m|}f'$ and $R''_{Em} = \big((m^2 - |m|)y^{|m|-2}\big)f + \big(2|m|y^{|m|-1}\big)f' + \big(y^{|m|}\big)f''$

$$\begin{aligned}
y^{|m|}f &= -\frac{1}{2\epsilon}\left[\frac{d^2}{dy^2} + \frac{1}{y}\frac{d}{dy} - \frac{m^2}{y^2} - y^2\right]y^{|m|}f \\
&= -\frac{1}{2\epsilon}\left[\left(\big((m^2 - |m|)y^{|m|-2}\big)f + \big(2|m|y^{|m|-1}\big)f' + \big(y^{|m|}\big)f''\right)\right. \\
&\qquad \left. + \frac{1}{y}\left(|m|y^{|m|-1}f + y^{|m|}f'\right) - \left(\frac{m^2}{y^2} + y^2\right)y^{|m|}f\right] \\
&= -\frac{y^{|m|}}{2\epsilon}\left[f'' + \left(\frac{2|m|}{y} + \frac{1}{y}\right)f' - \left(y^2\right)f\right] \\
f &= -\frac{1}{2\epsilon}\left[f'' + \left(\frac{2|m|}{y} + \frac{1}{y}\right)f' - \left(y^2\right)f\right]
\end{aligned}$$

On the second step, we will use $f = e^{-y^2/2}U$, and note that $f' = -ye^{-y^2/2}U + e^{-y^2/2}U'$ and $f'' = e^{-y^2/2}\Big[(y^2-1)U - 2yU' + U''\Big]$

$$\begin{aligned}
e^{-y^2/2}U &= -\frac{1}{2\epsilon}\Big[+ \left(e^{-y^2/2}\Big[(y^2-1)U - 2yU' + U''\Big]\right) \\
&\qquad + \left(\frac{2|m|}{y} + \frac{1}{y}\right)\Big(- ye^{-y^2/2}U + e^{-y^2/2}U'\Big) - \left(y^2\right)e^{-y^2/2}U\Big] \\
U &= -\frac{1}{2\epsilon}\left[U'' + \left(\frac{2|m|}{y} + \frac{1}{y} - 2y\right)U' - \Big(2|m| + 2\Big)U\right] \\
0 &= U'' + \left(\frac{2|m|+1}{y} - 2y\right)U' + \Big(2\epsilon - 2|m| - 2\Big)U
\end{aligned}$$

Part 6 Argue that a power series for U of the form

$$U(y) = \sum_{r=0}^{\infty} C_r y^r$$

will lead to a two term recursion relation

We plug in the power series into the previous step,

$$0 = \sum_{r=0}^{\infty} C_r(r^2 - r)y^{r-2} + \left(\frac{2|m|+1}{y} - 2y\right)\sum_{r=0}^{\infty} C_r r y^{r-1} + B\sum_{r=0}^{\infty} C_r y^r$$

Where we let $B \equiv 2\epsilon - 2|m| - 2$. Notice that the powers of y in the U' shift the terms up and down one order, so we can mix them into the other sums.

$$0 = \sum_{r=0}^{\infty} C_r\Big(r^2 + 2|m|r\Big)y^{r-2} + \sum_{r=0}^{\infty} C_r\Big(B - 2r\Big)y^r$$

Writing out a couple terms, we can see that the y^rth term, can be expressed as the two term relation.

$$\Big(C_{r+2}((r+2)^2 + 2|m|(r+2)) + C_r(B - 2r)\Big)y^r$$

Part 7 Find the relation between C_{r+2} and C_r. Argue that the series must terminate at some finite r if the $y \to \infty$ behavior of the solution is to be acceptable. Show $\epsilon = r + |m| + 1$ leads to termination after r terms. Now argue that r is necessarily even–i.e., $r = 2k$ (Show that if r is odd, the behavior of R as $\rho \to 0$ is not $\rho^{|m|}$.) So finally you end up with

$$E = (2k + |m| + 1)\hbar\omega, \qquad k = 0, 1, 2...$$

Define $n = 2k + |m|$, so that

$$E_n = (n+1)\hbar\omega$$

Since entire power series equals zero, we can equate the coefficient of the y^n to zero to arrive at a relation between C_{r+2} and C_r.

$$0 = C_{r+2}((r+2)^2 + 2|m|(r+2)) + C_r(B - 2r)$$
$$C_{r+2} = C_r\frac{2(|m| + 1 + r - \epsilon)}{(r+2)^2 + 2|m|(r+2)}$$

On physical grounds, we can argue that if $y \to \infty$, i.e., radially far from the isotropic oscillator, then the series should terminate at some finite r, otherwise we can see that the series U will not converge to a normalizable function.

See that at large r the recursion relation is approximately $C_{r+2} \approx C_r\frac{2r}{r^2} = C_r\frac{2}{r}$. Also consider the expansion $e^{y^2} = \sum_{n=0}^{\infty}\frac{y^{2n}}{n!}$ and how the coefficients a_n and a_{n+1} have a recursion relation $a_{n+1}/a_n = n!/(n+1)! = 1/(n+1) \approx 1/n$ for large n. Since the powers of y increase by two, we are also free to choose $n = r/2$ since n can be arbitrarily large. Then we see that $a_{r/2+1}/a_{r/2} \approx 2/r$ for large r.

Come back Zird I think this could be better explained. Particularly with the comparison to the exponential and the comparison of recursion relations.

The nonterminating series for $U(y)$ therefore has the dominant exponential growth as e^{y^2}. Consequentially, this makes radial equation $R = y^{|m|}e^{-y^2/2}U \approx e^{y^2}$, non normalizable leading to the contradiction.

If $\epsilon = r + |m| + 1$ then $C_{r+2} = 0$, i.e. C_r is the last term of the sequence.

We also can see that U contains only y^n terms where n is even. i.e. r is even.

Going back to the recursion expression in Part 6, we can write out a few terms

$$0 = C_0 B + C_1\left((1 + 2|m|\frac{1}{y} + (B-2)y\right) + ...$$

Notice that there is only one term wth a y^{-1} order, thus again, we equate it to zero to find that C_1, and hence all odd terms, are equal to zero. Then, U is left with only the even terms with a number of terms equal to $r = 2k$. We let $n \equiv 2k + |m|$

We can now rewrite

$$\begin{aligned} \epsilon &= r + |m| + 1 \\ E/\hbar\omega &= 2k + |m| + 1 \\ E &= (n+1)\hbar\omega \end{aligned}$$

Part 8 For a given n, what are the allowed values of $|m|$? Given this information show that for a given n, the degeneracy is $n + 1$. Compare this to what you found in cartesian coordinates

Let us illustrate a few examples. if $n = 0$, then $k = 0$ and $|m| = 0$. Thus we only have one possible state. if $n = 1$, then $k = 0$ but $|m| = 1$, since $m = \pm 1$ are both solutions then there is two possible states. $n = 2$ allows for $k = 1$, $|m| = 0$ or $k = 0$, $|m| = 2$, thus allowing for three possible states. Continued forward, we can see that the degeneracy is $n + 1$. This matches what was found earlier in cartesian coordinates.

Is m necessarily integer? Well come back to this Zird

Part 9 Write down all the normalized eigenfunctions corresponding to $n = 0, 1$.

the full normalized eigenfunction is the radial eigenfunction and the angular eigenfunction

$$\psi_{Em}(\rho, \phi) = R_{Em}(\rho)\Phi_m(\phi)$$

.

First we consider the lowest energy $n = 0$, which then sets our energy as $E = \hbar\omega$ and $\epsilon = 1$. This restricts $r = -|m|$, which is only possible if $|m| = 0$, i.e., our U function consist only of a constant C_0. . So $R_{\hbar\omega,0}(\rho) = e^{-\mu\omega\rho^2/2\hbar}C_0$. Additionally, the angular eigenfunciton is $\Phi_0(\phi) = \sqrt{\frac{1}{2\pi}}e^{i0\phi} = \sqrt{\frac{1}{2\pi}}$ Then

$$\psi_{\hbar\omega,0}(\rho, \phi) = \sqrt{\frac{1}{2\pi}}C_0 e^{-\mu\omega\rho^2/2\hbar}$$

.

C_0 can be determined by asserting the normalization condition

$$1 = \int_0^\infty \int_0^{2\pi} \left| \sqrt{\frac{1}{2\pi}} C_0 e^{-\mu\omega\rho^2/2\hbar} \right|^2 \rho d\rho d\phi = \frac{1}{2\pi}|C_0|^2 \int_0^{2\pi} d\phi \int_0^\infty \rho e^{-\mu\omega\rho^2/\hbar} d\rho$$
$$= |C_0|^2 \frac{\hbar}{2\mu\omega}$$

Thus solving for the normalization constant $C_0 = \sqrt{\frac{2\mu\omega}{\hbar}}$, the final eigenfunction for $n = 0$ is

$$\psi_{\hbar\omega,0} = \sqrt{\frac{\mu\omega}{\pi\hbar}} e^{-\mu\omega\rho^2/2\hbar}$$

If $n = 1$, the $E = 2\hbar\omega$. This restricts $r = 1 - |m| = 0$ for either case $m = \pm 1$. So again,

$$\psi_{2\hbar\omega,1} = R_{2\hbar\omega,1}(\rho)\Phi_{2\hbar\omega,1}(\phi) = \left(\rho e^{-\mu\omega\rho^2/2\hbar} C_0\right)\left(\sqrt{\frac{1}{2\pi}} e^{i\phi}\right)$$

Let's find the normalization constant. We make use of a gaussian integral that you can find online (or derive via substitution and IBP on A.2.2)

$$1 = \int_0^\infty \int_0^{2\pi} \left| \sqrt{\frac{1}{2\pi}} C_0 \rho e^{-\mu\omega\rho^2/2\hbar} e^{i\phi} \right|^2 \rho d\rho d\phi = \frac{|C_0|^2}{2\pi} \int_0^{2\pi} d\phi \int_0^\infty \rho^3 e^{-\mu\omega\rho^2/\hbar} d\rho$$
$$= \frac{|C_0|^2}{2\pi} \frac{\hbar^2}{2\mu^2\omega^2}$$

Thus $C_0 = \frac{2\mu\omega}{\hbar}\sqrt{\pi}$ and our final eigenfunction for $n = 1, m = 1$ is

$$\psi_{2\hbar\omega,1} = \sqrt{2}\frac{\mu\omega}{\hbar} \rho e^{-\mu\omega\rho^2/2\hbar} e^{i\phi}$$

Likewise, $\psi_{2\hbar\omega,-1} = \sqrt{2}\frac{\mu\omega}{\hbar} \rho e^{-\mu\omega\rho^2/2\hbar} e^{-i\phi}$

Part 10 Argue that the $n = 0$ function must equal the corresponding one found in Cartesian coordinates. Show that the two n=1 solutions are linear combinations of their counterparts in Cartesian coordinates. Verify that the parity of the states is $(-1)^n$ as you found in Cartesian coordinates.

Exercise 12.3.8 Consider a particle of mass μ and charge q in a vector potential

$$\vec{A} = \frac{B}{2}(-y\vec{i} + x\vec{j})$$

Part 1 Show that the magnetic field is $\vec{B} = B\vec{k}$

The magnetic field is given by the gradient of the potential

$$\vec{B} = \nabla \times \vec{A} = \frac{B}{2}(\partial_x x + \partial_y y)\vec{k} = B\vec{k}$$

Part 2 Show that a classical particle in this potential will move in circles at an angular frequency $\omega_0 = qB/\mu c$.

In cgs units, the lorentz force is as follows

$$\vec{F} = q(\vec{E} + \frac{\vec{v}}{c} \times \vec{B})$$

We can assume that the velocity is perpendicular from the magnetic field and that the electric field is zero. Equating this force to the form of uniform circular motion we get

$$\begin{aligned} F &= \frac{qvB}{c} = \frac{\mu v^2}{R} \\ \Rightarrow \frac{qB}{\mu c} &= \frac{v}{R} = \omega_0 \end{aligned}$$

where R is the radius of the circular path of the charged particle, and hence allows us to convert the velocity to an angular frequency.

Part 3 Consider the Hamiltonian for the corresponding quantum problem:

$$H = \frac{[P_x + qYB/2c]^2}{2\mu} + \frac{[P_y - qXB/2c]^2}{2\mu}$$

Show that $Q = (cP_x + qYB/2)/qB$ and $P = (P_y - qXB/2c)$ are canonical. Write H in terms of P and Q and show that allowed levels are $E_n = (n + 1/2)\hbar\omega_0$.

We have canonical transformation from X, Y, P_x, P_y to $Q(Y, P_x)$ and $P(X, P_y)$. To check whether they are canonical we can test them against the canonical equations

$$\begin{aligned} \{\bar{q}_j, \bar{q}_k\} &= 0 = \{\bar{p}_j, \bar{p}_k\} \\ \{\bar{q}_j, \bar{p}_k\} &= \delta_{jk} \end{aligned}$$

So it must be that $\{Q, P\} = 1$.

$$\{Q, P\} = \frac{\partial Q}{\partial X}\frac{\partial P}{\partial P_x} - \frac{\partial Q}{\partial P_x}\frac{\partial P}{\partial X} + \frac{\partial Q}{\partial Y}\frac{\partial P}{\partial P_y} - \frac{\partial Q}{\partial P_y}\frac{\partial P}{\partial Y} = \frac{1}{2} + \frac{1}{2} = 1$$

Now we can rewrite H.

$$\begin{aligned} H &= \frac{1}{2\mu}\left((Q\frac{qB}{c})^2 + P^2\right) \\ &= \frac{P^2}{2\mu} + \frac{1}{2}\mu\omega_0^2 Q^2 \end{aligned}$$

This hamiltonian takes the form of the 1D quantum harmonic oscillator with mass μ, angular frequency ω_0 and Q taking place of the typical X operator. Thus this hamiltonian has the same energy levels $E_n = (n + 1/2)\hbar\omega_0$

Part 4 Expand H out in terms of the original variables an show

$$H = H\big(\frac{\omega_0}{2}, \mu\big) - \frac{\omega_0}{2} L_z$$

where $H(\omega_0/2, \mu)$ is the Hamiltonian for an isotropic two-dimensional harmonic oscillator of mass μ and frequency $\omega/2$. Argue that the same basis that diagonalized $H(\omega_0/2, \mu)$ will diagonalize H. By thinking in terms of this basis, show that the allowed levels for H are $E = (k + \frac{1}{2}|m| - \frac{1}{2}m + \frac{1}{2})\hbar\omega_0$, where k is any integer (0,1,2,...) and m is the angular momentum. Convince yourself that you get the same levels from this formula as from the earlier one.

. . . . . .

By Theorem 13, if two Hermitian operators commute, then there exists a basis of common eigenvectors that diagonalized them both.

So this amounts to showing that the obviously Hermitian operators H and $H(\omega_0/2, \mu)$ commute. Let $H_{1/2} \equiv H\big(\frac{\omega_0}{2}, \mu\big)$

$$[H, H_{1/2}] = [H_{1/2} - \frac{\omega_0}{2} L_z, H_{1/2}] = [H_{1/2}, H_{1/2}] - \frac{\omega_0}{2}[L_z, H_{1/2}] = 0$$

The above step is clear since this is a rotationally invariant problem, i.e. $[L_z, H_{1/2}] = 0$.

Now, if we use this basis to diagonalize $H_{1/2}$, then since it is two dimensional harmonic oscillator with mass μ and angular frequency ω_2, it would have energy eigenvalues $E = (2k + |m| + 1)\hbar\frac{\omega_0}{2}$

Similarly, if we use this basis to diagonalize $-\frac{\omega_0}{2} L_z$, we get eigenvalues $-\frac{\omega_0}{2} m\hbar$. Adding these two diagonal matrices gives us the energy eigenvalues for the full H $E = (k + \frac{1}{2}|m| - \frac{1}{2}m + \frac{1}{2})\hbar\omega_0$.

Notice that for $m > 0$ the m terms cancel such that $E = (k + \frac{1}{2})\hbar\omega_0$ but if $m < 0$ then the m terms add such that $E = (k + |m| + \frac{1}{2})\hbar\omega_0$. But expressions are functionally equivalent to $E = (n + \frac{1}{2})\hbar\omega_0$ since k or $k + |m|$ are both some integer.

Exercise 12.4.1

Part 1 Verify that $\vec{c} = \vec{a} \times \vec{b}$ is equivalent to $c_i = \sum_{j=1}^{3} \sum_{k=1}^{3} \epsilon_{ijk} a_j b_k$, given the definition of ϵ_{ijk}

. . . . . .

We can expand the cross product to see

$$\vec{c} = (a_2 b_3 - a_3 b_2)\vec{i} - (a_1 b_3 - a_3 b_1)\vec{j} + (a_1 b_2 - a_2 b_1)\vec{k}$$

Each coefficient of the vectors can be verified using the ϵ_{ijk} tensor.

$$\begin{aligned} c_1 &= \epsilon_{123} a_2 b_3 + \epsilon_{132} a_3 b_2 = a_2 b_3 - a_3 b_2 \\ c_2 &= \epsilon_{213} a_1 b_3 + \epsilon_{231} a_3 b_1 = a_3 b_1 - a_1 b_3 \\ c_3 &= \epsilon_{312} a_1 b_2 + \epsilon_{321} a_2 b_1 = a_1 b_2 - a_2 b_1 \end{aligned}$$

Part 2 Let U_1, U_2, U_3 be three energy eigenfunctions of a single particle in some potential, Construct the wave function $\psi_A(x_1, x_2, x_3$ of three fermions in this potential, one of which is in U_1, one in U_2, and one in U_3, using the ϵ_{ijk} tensor.

The wavefunction for fermions (which are antisymmetric) is according to equation 10.3.35

$$\begin{aligned}\psi_A(x_1, x_2, x_3) = 6^{-1/2}\big(&U_1(x_1)U_2(x_2)U_3(x_3) - U_1(x_1)U_3(x_2)U_3(x_3)\\ &U_2(x_1)U_3(x_2)U_1(x_3) - U_2(x_1)U_1(x_2)U_3(x_3)\\ &U_3(x_1)U_1(x_2)U_2(x_3) - U_3(x_1)U_2(x_2)U_3(x_3)\big)\end{aligned}$$

This can be condensed as

$$\psi_A(x_1, x_2, x_3) = \frac{1}{\sqrt{6}}\sum_{i=1}^{3}\sum_{j=1}^{3}\sum_{k=1}^{3}\epsilon_{ijk}U_i(x_1)U_j(x_2)U_k(x_3)$$

Exercise 12.4.2

Part 1 Verify Eq 12.4.2 by first constructing $3x3$ matrices corresponding to $\mathbb{R}(\epsilon_x\vec{i})$ and $\mathbb{R}(\epsilon_y\vec{j})$, to order ϵ.

$$\mathbb{R}(-\epsilon_y\vec{j})\mathbb{R}(-\epsilon_x\vec{i})\mathbb{R}(\epsilon_y\vec{j})\mathbb{R}(\epsilon_x\vec{i}) = \mathbb{R}(-\epsilon_x\epsilon_y\vec{k})$$

Using the rotation matrices and approximations of order ϵ, this corresponds to

$$\begin{bmatrix}1 & 0 & -\epsilon_y\\ 0 & 1 & 0\\ \epsilon_y & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & \epsilon_x\\ 0 & -\epsilon_x & 1\end{bmatrix}\begin{bmatrix}1 & 0 & \epsilon_y\\ 0 & 1 & 0\\ -\epsilon_y & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & -\epsilon_x\\ 0 & \epsilon_x & 1\end{bmatrix} = \begin{bmatrix}1 & \epsilon_x\epsilon_y & 0\\ -\epsilon_x\epsilon_y & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

Carrying out the matrix multiplication gives on the LHS

$$\begin{bmatrix}1 & \epsilon_x\epsilon_y & -\epsilon_y\\ 0 & 1 & \epsilon_x\\ \epsilon_y & -\epsilon_x & 1\end{bmatrix}\begin{bmatrix}1 & \epsilon_x\epsilon_y & \epsilon_y\\ 0 & 1 & -\epsilon_x\\ -\epsilon_y & \epsilon_x & 1\end{bmatrix} = \begin{bmatrix}1+\epsilon_y^2 & \epsilon_x\epsilon_y & -\epsilon_x^2\epsilon_y\\ -\epsilon_x\epsilon_y & 1+\epsilon_x^2 & 0\\ 0 & \epsilon_x\epsilon_y^2 & 1+\epsilon_x^2+\epsilon_y^2\end{bmatrix}$$

Which after getting rid of higher order terms, the LHS matches the RHS.

Part 2 Provide the steps connecting the quantum operators version of this with $U[\mathbb{R}]$ to $[L_x, L_y] = i\hbar L_z$

The quantum operator version can be evaluated below

$$\big(\mathbb{1} + \frac{i}{\hbar}\epsilon_y L_y\big)\big(\mathbb{1} + \frac{i}{\hbar}\epsilon_x L_x\big)\big(\mathbb{1} - \frac{i}{\hbar}\epsilon_y L_y\big)\big(\mathbb{1} - \frac{i}{\hbar}\epsilon_x L_x\big) = \big(\mathbb{1} + \frac{i}{\hbar}\epsilon_x\epsilon_y L_z\big)$$

On the LHS we can evaluate the product and only consider terms of order $\mathbb{1}$ and $\epsilon_x\epsilon_y$ for the purpose of matching coefficients with the RHS.

$$\begin{aligned}&\Big(\mathbb{1}+\frac{i}{\hbar}(\epsilon_x L_x+\epsilon_y L_y)-\frac{1}{\hbar^2}\epsilon_x\epsilon_y L_y L_x\Big)\Big(\mathbb{1}-\frac{i}{\hbar}(\epsilon_x L_x+\epsilon_y L_y)-\frac{1}{\hbar^2}\epsilon_x\epsilon_y L_y L_x\Big)\\ =&\mathbb{1}+\frac{\epsilon_x\epsilon_y}{\hbar^2}\Big[L_xL_y-L_yL_x\Big]+O(\epsilon_x,\epsilon_y,\epsilon_x^2\epsilon_y,\epsilon_x\epsilon_y^2,\epsilon_x^2,\epsilon_y^2)\end{aligned}$$

Thus equating coefficients we get

$$\frac{i}{\hbar}L_z=\frac{1}{\hbar^2}[L_x,L_y]\Rightarrow[L_x,L_y]=i\hbar L_z$$

Part 3 Verify that L_x and L_y satisfy the above relation. The proof for other commutators follows by cyclic permutation.

We have definition $L_x=YP_z-ZP_y$ and $L_y=ZP_x-XP_z$, so we see that,

$$\begin{aligned}[L_x,L_y]&=(YP_z-ZP_y)(ZP_x-XP_z)-(ZP_x-XP_z)(YP_z-ZP_y)\\&=YP_zZP_x-YP_zXP_z-ZP_yZP_x+ZP_yXP_z\\&-(ZP_xYP_z-ZP_xZP_y-XP_zYP_z+XP_zZP_y)\end{aligned}$$

Remember that $[X_i,P_j]=i\hbar\delta_{ij}$, and $[X_i,X_j]=[P_i,P_j]=0$. This allows us to cancel some terms

$$\begin{aligned}&=YP_zZP_x+ZP_yXP_z-ZP_xYP_z-XP_zZP_y\\&=Z(P_yX-P_xY)P_z+P_zZ(YP_x-XP_y)\\&=[Z,P_z]L_z=i\hbar L_z\end{aligned}$$

Exercise 12.4.3 We would like to show that $\hat{\theta}\cdot\vec{L}$ generates rotation about the axis parallel to $\hat{\theta}$. Let $\delta\vec{\theta}$ be an infinitesimal rotation parallel to $\vec{\theta}$.

Part 1 Show that when a vector $\vec{r}$ is rotated by an angle $\delta\vec{\theta}$, it changes to $\vec{r}+\delta\vec{\theta}\times\vec{r}$. (It might help to start with $\vec{r}\perp\delta\vec{\theta}$ and then generalize)

Firstly, this is evident from drawing a diagram. Because the part of $\vec{r}\parallel\hat{\theta}$ is not affected a rotation along $\hat{\theta}$ and the perpendicular part would rotate an infinitesimal arc length along $\hat{\phi}$ of magnitude $r\delta\theta$

More concretely,

$$\vec{r}=\vec{r}_{\perp\theta}+\vec{r}_{\parallel\theta}\xrightarrow[R(\delta\theta)]{}\vec{r}_{\perp\theta}+r\sin(\delta\theta)(\hat{\theta}\times\hat{r})+\vec{r}_{\parallel\theta}=\vec{r}+\delta\vec{\theta}\times\vec{r}$$

Part 2 We therefore demand that (to first order, as usual). Note that this is just a Taylor expansion of ψ

$$\psi(\vec{r}) \xrightarrow[U(\mathbb{R}(\delta\vec{\theta}))]{} \psi(\vec{r} - \delta\vec{\theta} \times \vec{r}) = \psi(\vec{r}) - (\delta\vec{\theta} \times \vec{r}) \cdot \nabla\psi$$

Comparing to $U(\mathbb{R}(\delta\vec{\theta})) = \mathbb{1} - \frac{i}{\hbar}\delta\theta L_{\hat{\theta}}$, show that $L_{\hat{\theta}} = \hat{\theta} \cdot \vec{L}$

The comparison is more clear by applying $U(\mathbb{R}(\delta\vec{\theta}))$ to an abstract ψ.

$$U(\mathbb{R}(\delta\vec{\theta}))\,|\psi\rangle = \big(\mathbb{1} - \frac{i}{\hbar}\delta\theta L_{\hat{\theta}}\big)\,|\psi\rangle = |\psi\rangle - \frac{i}{\hbar}\delta\theta L_{\hat{\theta}}\,|\psi\rangle$$

Now it is clear that in the coordinate basis, then the second term of the expansion is equivalent to the second term in the abstract.

Let's take a closer look at the Taylor expansion's second term. We will make use of a scalar cross product identities and also remember that since in the coordinate basis $\vec{p} = -i\hbar\vec{\nabla}$, then $\vec{L} = \vec{r} \times \vec{p} = -i\hbar(\vec{r} \times \nabla)$

$$-(\delta\vec{\theta} \times \vec{r}) \cdot \nabla = -\delta\vec{\theta} \cdot (\vec{r} \times \nabla) = \frac{1}{i\hbar}(\delta\vec{\theta} \cdot \vec{L}) = -\frac{i}{\hbar}\delta\theta(\hat{\theta} \cdot \vec{L})$$

Now the two matches form and it is clear that $L_{\hat{\theta}} = \hat{\theta} \cdot \vec{L}$

Exercise 12.4.4 Recall that $\vec{V}$ is a vector operator if its components V_i transform as

$$U^\dagger[R]V_iU[R] = \sum_j R_{ij}V_j$$

Part 1 For an infinitesimal rotation $\delta\vec{\theta}$, show, on basis of the previous exercise, that

$$\sum_j R_{ij}V_j = V_i + (\delta\vec{\theta} \times \vec{V})_i = V_i + \sum_j \sum_k \epsilon_{ijk}(\delta\theta)_j V_k$$

The LHS shows a component of vector $\vec{V}$ transforming under some small rotation $R[\delta\vec{\theta}]$. Then according to the last exercise, $V_i + (\delta\vec{\theta} \times \vec{V})_i$ is just the single-component view of how V is rotated. The last expression follows from emrely the definition of cross products with ϵ_{ijk} (as seen in eq. 12.4.8 & 12.4.9)

Part 2 Feed in $U[R] = 1 - (i/\hbar)\delta\vec{\theta} \cdot \vec{L}$ into the left-hand side of the initial transformation equation and deduce that

$$[V_i, L_j] = i\hbar \sum_k \epsilon_{ijk}V_k$$

$$\begin{aligned} U^\dagger V_i U &= \big(1 + \frac{i}{\hbar}\delta\vec{\theta} \cdot \vec{L}\big)V_i\big(1 + \frac{i}{\hbar}\delta\vec{\theta} \cdot \vec{L}\big) \\ &= V_i + \frac{i}{\hbar}\Big[(\delta\vec{\theta} \cdot \vec{L})V_i - V_i(\delta\vec{\theta} \cdot \vec{L})\Big] + \frac{1}{\hbar^2}(\delta\vec{\theta} \cdot \vec{L})V_i(\delta\vec{\theta} \cdot \vec{L}) \end{aligned}$$

We will ignore the third term because it will contain terms of order $(\delta\theta)_j^2$ which is too much for our first order expansion of U. We can now equate this to the answer to part 1, cancel out V_i, and continue to evaluate

$$\frac{i}{\hbar}\left[\sum_j (\delta\theta)_j L_j V_i - V_i (\delta\theta)_j L_j\right] = \sum_{jk} \epsilon_{ijk}(\delta\theta)_j V_k$$

To match the form of the desired commutation, match coefficients $(\delta\theta)_j$ across the left and right hand sides

$$(\delta\theta)_j L_j V_i - V_i(\delta\theta)_j L_j = -i\hbar \sum_k \epsilon_{ijk}(\delta\theta)_j V_k$$
$$[V_i, L_j] = i\hbar \sum_k \epsilon_{ijk} V_k$$

This is as good a definition of a vector operator as the first definition. By setting $\vec{V} = \vec{L}$, we can obtain the commutation rules among the L's.

Exercise 12.5.1 Consider a vector field $\vec{\Psi}(x, y)$ in two dimensions. From fig 12.1 it follows that under an infinitesimal rotation $\epsilon_z \vec{k}$,

$$\Psi_x \to \Psi'_x(x, y) = \Psi_x(x + y\epsilon_z, y - x\epsilon_z) - \Psi_y(x + y\epsilon_z, y - x\epsilon_z)\epsilon_z$$
$$\Psi_y \to \Psi'_y(x, y) = \Psi_x(x + y\epsilon_z, y - x\epsilon_z)\epsilon_z + \Psi_y(x + y\epsilon_z, y - x\epsilon_z)$$

Show that (to order ϵ_z)

$$\begin{bmatrix}\Psi'_x \\ \Psi'_y\end{bmatrix} = \left(\begin{bmatrix}1 & 0 \\ 0 & 1\end{bmatrix} - \frac{i\epsilon_z}{\hbar}\begin{bmatrix}L_z & 0 \\ 0 & L_z\end{bmatrix} - \frac{i\epsilon_z}{\hbar}\begin{bmatrix}0 & -i\hbar \\ i\hbar & 0\end{bmatrix}\right)\begin{bmatrix}\Psi_x \\ \Psi_y\end{bmatrix}$$

so that

$$J_z = L_z^{(1)} \otimes I^{(2)} + I^{(1)} \otimes S_z^{(2)} = L_z + S_z$$

where $I^{(2)}$ is a 2x2 identity matrix with respect to the vector components, $I^{(1)}$ is the identity operator with respect to the argument (x, y) of $\vec{\Psi}(x, y)$. This example only illustrates the fact that $J_z = L_z + S_z$ if the wavefunction is not a scalar. An example of half-integral eigenvalues will be provided when we consider spin in a later chapter. (In the present example, S_z has eigenvalues $\pm\hbar$

Let us write the first rotations in the abstract

$$|\Psi'_x\rangle = U[\mathbb{R}(\epsilon_z\vec{k})]\,|\Psi_x\rangle - U[\mathbb{R}(\epsilon_z\vec{k})]\epsilon_z\,|\Psi_y\rangle$$
$$= \mathbb{1}\,|\Psi_x\rangle - \frac{i\epsilon_z}{\hbar}L_z\,|\Psi_x\rangle - \epsilon_z\,|\Psi_y\rangle - O(\epsilon_z^2)$$
$$\xrightarrow[x,y,basis]{} \Psi'_x = \Psi_x - \frac{i\epsilon_z}{\hbar}L_z\Psi_x - \epsilon_z\Psi_y$$

Similarly,

$$\Psi'_y = \Psi_y - \frac{i\epsilon_z}{\hbar} L_z \Psi_y + \epsilon_z \Psi_x$$

Combining these two equations into one matrix equation, yields exactly the matrix equation needed. The relation with J_z is demonstrated in that $L_z^{(1)}$ acts as normal and would generate the infinitesimal translation of the argument (x, y). Likewise, $S_z^{(2)}$ acts on the vector components as it introduces cross-terms to the expressions for Ψ'_x and Ψ'_y.

Exercise 12.5.2

Part 1 Verify that the 2 x 2 matrices $J_x^{(1/2)}$, $J_y^{(1/2)}$, $J_z^{(1/2)}$ obey the commutation rule $[J_x^{(1/2)}, J_y^{(1/2)}] = i\hbar J_z^{(1/2)}$

We write out the matrices

$$\begin{aligned}
[J_x^{(1/2)}, J_y^{(1/2)}] &= \begin{bmatrix} 0 & \hbar/2 \\ \hbar/2 & 0 \end{bmatrix} \begin{bmatrix} 0 & -i\hbar/2 \\ i\hbar/2 & 0 \end{bmatrix} - \begin{bmatrix} 0 & -i\hbar/2 \\ i\hbar/2 & 0 \end{bmatrix} \begin{bmatrix} 0 & \hbar/2 \\ \hbar/2 & 0 \end{bmatrix} \\
&= i\hbar^2/4 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + i\hbar^2/4 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\
&= i\hbar \begin{bmatrix} \hbar/2 & 0 \\ 0 & -\hbar/2 \end{bmatrix} = i\hbar J_z^{(1/2)}
\end{aligned}$$

Part 2 Do the same for the 3 x 3 matrices $J_i^{(1)}$

$$[J_x^{(1)}, J_y^{(2)}] = \frac{\hbar}{\sqrt{2}} \frac{i\hbar}{\sqrt{2}} \left(\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} - \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \right)$$

$$= \frac{i\hbar^2}{2} \left(\begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{bmatrix} - \begin{bmatrix} -1 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \right) = i\hbar \begin{bmatrix} \hbar & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\hbar \end{bmatrix} = i\hbar J_z^{(1)}$$

Part 3 Construct the 4 x 4 matrices and verify that

$$[J_x^{(3/2)}, J_y^{3/2}] = i\hbar J_z^{(3/2)}$$

Don't waste ur time with this Zird

Exercise 12.5.3

Part 1 Show that $\langle J_x \rangle = \langle J_y \rangle = 0$ in a state $|jm\rangle$.

$$\langle jm| J_x |jm\rangle = \frac{\hbar}{2} \{ \delta_{jj} \delta_{m,m+1} \sqrt{(j-m)(j+m+1)} + \delta_{jj} \delta_{m,m-1} \sqrt{(j+m)(j-m+1)} \} = 0$$

Because $\delta_{m,m\pm1}$ will never be non-zero, the above is clear. Similarly for $\langle J_y \rangle = 0$

Part 2 Show that in these states

$$\langle J_x^2 \rangle = \langle J_y^2 \rangle = \frac{1}{2}\hbar^2[j(j+1) - m^2]$$

We can expand J_x^2 with $J_x = (J_+ + J_-)/2$

$$\langle J_x^2 \rangle = \frac{1}{4}\big[\, \langle J_+^2 \rangle + \langle J_-^2 \rangle + \langle J_+ J_- \rangle + \langle J_- J_+ \rangle \,\big]$$

For similar reasons as above, $\langle J_\pm^2 \rangle = 0$ because they will produce $\delta_{m,m\pm 2}$ terms that will never be zero. Then,

$$\begin{aligned}
\langle J_x^2 \rangle &= \frac{1}{4}\Big[\langle jm|\, J_+ C_-(j,m)\, |j, m-1\rangle + \langle jm|\, J_- C_+(j,m)\, |j, m+1\rangle \Big] \\
&= \frac{1}{4}\Big[C_+(j,m-1)C_-(j,m) + C_-(j,m+1)C_+(j,m) \Big] \\
&= \frac{1}{4}\Big[\hbar^2(j+m)(j-m+1) + \hbar^2(j-m)(j+m+1) \Big] \\
&= \frac{\hbar^2}{2}\Big[j(j+1) - m^2 \Big]
\end{aligned}$$

$\langle J_y^2 \rangle$ follows the exact same solution because

$$J_y^2 = \langle [(J_+ - J_-)/2i]^2 \rangle = \langle -(J_+^2 + J_-^2 - J_+J_- - J_-J_+)/4 \rangle = \langle J_x^2 \rangle$$

Part 3 Check that $\Delta J_x \cdot \Delta J_y$ from part (2) satisfies the inequality imposed by the uncertainty principle

$$(\Delta J_x)^2 (\Delta J_y)^2 \geq |\, \langle \hat{J}_x \hat{J}_y \rangle \,|^2$$

First, we note that because $\langle J_x \rangle = 0$, then $\hat{J}_x = J_x - \langle J_x \rangle = J_x$, and the same for $\hat{J}_y$. This also allows for our uncertainty $(\Delta J_x)^2 = \langle \hat{J}_x^2 \rangle = \langle J_x^2 \rangle$, and $(\Delta J_y)^2 = \langle J_y^2 \rangle$

Plugging in our solutions for the LHS, we get

$$\left(\frac{\hbar^2}{2}[j(j+1) - m^2] \right)^2 = \frac{\hbar^4}{4}\Big[j^4 + 2j^3 + j^2 - 2m^2 j^2 - 2jm^2 + m^4 \Big]$$

The RHS can be expanded with $J_\pm$

$$\begin{aligned}
\left| \langle \hat{J}_x \hat{J}_y \rangle \right|^2 = \left| \langle J_x J_y \rangle \right|^2 &= \left| \frac{1}{4i}\Big[\langle J_+^2 \rangle - \langle J_-^2 \rangle + \langle J_- J_+ \rangle - \langle J_+ J_- \rangle \Big] \right|^2 \\
&= \left| \frac{1}{4i}\Big[\hbar^2(j-m)(j+m+1) - \hbar^2(j+m)(j-m+1) \Big] \right|^2 \\
&= |-m\hbar^2/2i|^2 = (\hbar^4/4)(m^2)
\end{aligned}$$

We now remember that $|m| \leq j$ for all their values. This also means that $j^2 - m^2 \geq 0$. This will be useful in showing the inequality.

$$\frac{\hbar^4}{4}\left[j^4+2j^3+j^2-2m^2j^2-2jm^2+m^4\right] \geq \frac{\hbar^2}{4}m^2$$
$$(j^4-2j^2m^2+m^4)+(2j^3-2jm^2)+(j^2-m^2) \geq 0$$
$$(j^2-m^2)^2+2j(j^2-m^2)+(j^2-m^2) \geq 0$$

All terms are ≥ 0 so the inequality is satisfied.

Part 4 Show that the uncertainty bound is saturated in the state $|j,\pm j\rangle$.

This is clear as such a scenario would assert that $j^2-m^2=0$, hence the inequality is saturated with $0 \geq 0$

Exercise N/A Exercises 12.5.4 to 12.5.7 are skipped, as the Finite rotations section is to be skipped in a "less advanced course". 12.5.8 is also skipped as it is optional.

Come back to this Zird

Exercise 12.5.9 Show that L^2 above is Hermitian in the sense

$$\int \psi_1^*(L^2\psi_2)d\Omega = \left[\int \psi_2^*(L^2\psi_1)d\Omega\right]^*$$

Let's start on the LHS. We use the coordinate basis of L^2 (eq. 12.5.36) and the solid angle formula $d\Omega = \sin\theta d\phi d\theta$

$$\begin{aligned}\int \psi_1^*(L^2\psi_2)d\Omega &= \int \psi_1^*\left[-\hbar^2\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta}+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right)\right]\psi_2\sin\theta d\phi d\theta \\ &= -\hbar^2\left[\int \psi_1^*\frac{\partial}{\partial\theta}\left[\sin\theta\frac{\partial\psi_2}{\partial\theta}\right]d\phi d\theta + \int \frac{1}{\sin\theta}\psi_1^*\frac{\partial^2\psi_2}{\partial\phi^2}d\phi d\theta\right]\end{aligned}$$

We take note that generally, by applying integral by parts once or twice, we can evaluate,

$$\int_a^b f(x)g'(x)dx = f(x)g(x)\Big|_a^b - \int_a^b f'(x)g(x)dx$$
$$\int_a^b f(x)g''(x)dx = f(x)g'(x)\Big|_a^b - f'(x)g(x)\Big|_a^b + \int_a^b f''(x)g(x)dx$$

Provided that the boundary terms are zero (often for periodic functions on the limits of integration), then essentially the derivatives can be interchanged between the two functions.

For the first integral, we transfer the first $\partial\theta$ derivative onto the ψ_1^* and transfer ψ_2's derivative onto $\frac{\partial\psi_1^*}{\partial\theta}\sin\theta$. The boundary term is handled by a $\sin\theta$ factor that integrates to zero from $[0,\pi]$. The second term, integrated under ϕ now, simply can transfer the full second derivative from ψ_2 to ψ_1^*. The boundary term is taken care of since ψ_1 and ψ_2, and all their derivatives take the same value on boundaries $\phi=[0,2\pi]$. Using these formulas lets us right the LHS in the form of the RHS after conjugation

$$= -\hbar^2\left[\int \psi_2^*\frac{\partial}{\partial\theta}\left[\sin\theta\frac{\partial\psi_1}{\partial\theta}\right]d\phi d\theta + \int \frac{1}{\sin\theta}\psi_2^*\frac{\partial^2\psi_1}{\partial\phi^2}d\phi d\theta\right] = \left[\int \psi_2^*(L^2\psi_1)d\Omega\right]^*$$

Exercise 12.5.10

Part 1 Write the differential equation corresponding to

$$L^2 |\alpha\beta\rangle = \alpha |\alpha\beta\rangle$$

in the coordinate basis, using the L^2 operator given in Eq. (12.5.36). We already know $\beta = m\hbar$ from the analysis of $-i\hbar(\partial/\partial\phi)$. So assume that the simultaneous eigenfunctions have the form

$$\psi_{\alpha m}(\theta, \phi) = P_\alpha^m(\theta) e^{im\phi}$$

and show that P_α^m satisfies the equation

$$\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta} + \frac{\alpha}{\hbar^2} - \frac{m^2}{\sin^2\theta}\right)P_\alpha^m(\theta) = 0$$

. . . . . .

See that,

$$L^2 |\alpha\beta\rangle = \alpha |\alpha\beta\rangle$$
$$-\hbar^2\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right)\psi_{\alpha\beta}(\theta,\phi) = \alpha\psi_{\alpha\beta}(\theta,\phi)$$
$$\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left[\sin\theta\frac{\partial}{\partial\theta}\right] + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2} + \frac{\alpha}{\hbar^2}\right)P_\alpha^m(\theta)e^{im\phi} = 0$$
$$\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left[\sin\theta\frac{\partial}{\partial\theta}\right] + \frac{\alpha}{\hbar^2} - \frac{m^2}{\sin^2\theta}\right)P_\alpha^m(\theta) = 0$$

Part 2 We need to show that

$$(1) \qquad \frac{\alpha}{\hbar^2} = l(l+1), \qquad l = 0, 1, 2...$$
$$(2) \qquad |m| \leq l$$

We will consider only part (1) and that too for the case m=0. By rewriting the equation in terms of $u = \cos\theta$, Show that P_α^0 satisfies

$$(1-u^2)\frac{d^2P_\alpha^0}{du^2} - 2u\frac{dP_\alpha^0}{du} + \frac{\alpha}{\hbar^2}P_\alpha^0 = 0$$

. . . . . .

The variable change is accompanied by the following substitutions: $(1-u^2) = \sin^2\theta$, and $d/d\theta = (d/du)(du/d\theta) = -\sin\theta(d/du)$. Pay careful attention that,

$$\frac{d^2}{d\theta^2} = \sin\theta\left(-\cos\theta\frac{1}{\sin\theta}\frac{d}{du} - \sin\theta\frac{d^2}{du^2}\right) = -u\frac{d}{du} + (1-u^2)\frac{d^2}{du^2} \neq (1-u^2)\frac{d^2}{du^2}$$

Applying these to the equation, we note that the $\alpha/\hbar^2$ term is preserved and the m^2 term is zero since $m = 0$ in this particular case. All that is left is to take the derivative using the product rule and substitute the derivatives, the final form follows directly from that substitution.

$$\left(\frac{1}{\sin\theta}\left[\cos\theta P_\alpha^{\prime m}(\theta) + \sin\theta P_\alpha^{\prime\prime m}(\theta)\right] + \frac{\alpha}{\hbar^2}P_\alpha^m\right) = 0$$
$$(1-u^2)\frac{d^2P_\alpha^0}{du^2} - 2u\frac{dP_\alpha^0}{du} + \frac{\alpha}{\hbar^2}P_\alpha^0 = 0$$

Part 3 Convince yourself that a power series solution

$$P_\alpha^0 = \sum_{n=0}^{\infty} C_n u^n$$

will lead to a two term recursion relation. Show that $(C_{n+2}/C_n) \to 1$ as $n \to \infty$. Thus the series diverges when $|u| \to 1$ ($\theta \to 0$ or π). Show that if $\alpha/\hbar^2 = l(l+1), \quad l = 0, 1, 2...$, the series will terminate and be either an even or odd function of u. The functions $P_\alpha^0 = P_{l(l+1)\hbar^2}^0 \equiv P_l^0 \equiv P_l$ are just the Legendre polynomials up to a scale factor. Determine P_0, P_1, and P_2 and compare (ignoring overall scales) with the Y_l^0 funciton.

We start by writing out the derivatives of $P_\alpha^0(u)$

$$P_\alpha^{\prime 0}(u) = \sum_{n=0}^{\infty} C_n n u^{n-1}$$
$$P_\alpha^{\prime\prime 0}(u) = \sum_{n=0}^{\infty} C_n n(n-1) u^{n-2}$$

Plugging these into the previous equation we get

$$(1-u^2)\sum_{n=0}^{\infty} C_n n(n-1)u^{n-2} - 2u\sum_{n=0}^{\infty} C_n n u^{n-1} + \frac{\alpha}{\hbar^2}\sum_{n=0}^{\infty} C_n u^n = 0$$

Lets observe the coefficient for the term of u^x, which should equal zero when compared to the RHS. Observe that after multiplying out the u terms, we get a two term relation

$$C_{x+2}(x+2)(x+1) - C_x x(x-1) - 2C_x x + \frac{\alpha}{\hbar^2} C_x = 0$$

Observe that as $x \to \infty$, $C_{x+2}/C_x \to 1$

$$C_{x+2}/C_x = \frac{x(x-1) + 2x - \frac{\alpha}{\hbar^2}}{(x+2)(x+1)} \xrightarrow[x\to\infty]{} \approx \frac{x^2}{x^2} = 1$$

Because of that,

$$P_\alpha^0(u) \xrightarrow[x\to\infty]{} \sum_{n=0}^{\infty} C_x |u|$$

which will not converge if $|u| \to 1$.

If we allow that $\alpha/\hbar^2 = l(l+1)$, the $C_{x+2}/C_x = 0$ if $x = l$, i.e. if the series terminates after l terms. However, the recursion relation places no restriction on coefficients with opposite parity of l. But on physical grounds, we insist that all these coefficients must equal zero. If not, then the power series of P will not terminate and lead to non physical solutions to this differential equation. Hence, the parity of l completely determines the parity of P.

Another way of viewing this is that we have to select a value for two constant from the second order differential equations: C_0 and C_1, each of them sets in stone the coefficients of the same parity for the series according to the recursion relation. If $\alpha/\hbar^2 = l(l+1)$ then the coefficients with the same parity of l will terminate at C_l. However, for any non-zero choice of the constant with opposite parity (either C_0 or C_1, then the recursion relation will never terminate, and cause an un-physical solution. Thus we are left with the only choice of choosing zero for the opposite parity coefficients.

We now say that $P_l = C_l u^l + C_{l-2}u^{l-2} + ...$ ending at either C_0 or C_1 depending on the parity of l. Thus $P_0 = C_0$, $P_1 = C_1 \cos\theta$ and $P_2 = C_0 + C_2 \cos^2\theta$. On the last one, it can be determined via the two term recursion that $C_2 = -3C_0$ so really $P_2 = C_0(1 - 3\cos^2\theta)$. These, ignoring overall scales, match the form of their respective Y_l^0 functions.

Exercise 12.5.11 Derive Y_1^1 starting from Eq. (12.5.28) and normalize it yourself, remembering the phase convention. Lower it to get Y_1^0 and Y_1^-1 and compare.

Starting from

$$\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi}\right)\psi_l^l(r,\theta,\phi) = 0$$

Since ψ_l^l is an eigen function of L_z with eigenvalue of $l\hbar$, we can let $\psi_l^l(r,\theta,\phi) = U_l^l(r,\theta)e^{i\phi}$ which turns our equation to

$$\left(\frac{\partial}{\partial\theta} - l\cot\theta\right)U_l^l = 0$$

This can be integrated so that $U_l^l(r,\theta) = R(r)\sin^l\theta$ which gives wave function

$$\psi_l^l(r,\theta,\phi) = R(r)\sin^l\theta e^{i\phi}$$

All of this follows from the textbook. Now we let the angular part of this equation be called $Y_l^l = \sin^l\theta e^{i\phi}$. Which is the unique, non-degenerate solution in the absence of the radial coordinate. This means that it should obey the following normalization condition. We append a factor A for this to be determine normalization factor From hence forth we will now restrict our analysis to where $l = 1$

$$\begin{aligned}
1 &= \int |Y_1^1|^2 \sin\theta d\theta d\phi \\
&= \int |A|^2 \sin^3\theta e^{\phi(i-i)} d\theta d\phi = |A|^2 \Big[\phi\Big]_0^{2\pi} \int_0^\pi d\theta \sin\theta(1-\cos^2\theta) \\
&= |A|^2 2\pi \left[-\cos\theta + \frac{1}{3}\cos^3\theta\right]\Bigg|_0^\pi = |A|^2 2\pi(2-2/3) = |A|^2 8\pi/3
\end{aligned}$$

If we consider the $(-1)^l$ phase convention, we get,

$$Y_1^1(\theta,\phi) = -\sqrt{\frac{3}{8\pi}}\sin\theta e^{i\phi}$$

We can get Y_l^0 using the lowering operator L_- in the coordinate basis. We recall its eigenvalues acting on its eigenfunction follow $J_- |lm\rangle = \hbar[(j+m)(j-m+1)]^{1/2} |j, m-1\rangle$

$$\begin{aligned} L_- Y_1^1(\theta,\phi) = \hbar\sqrt{2}Y_1^0(\theta,\phi) &= -\hbar e^{-i\phi}\left(\frac{\partial}{\partial\theta} - i\cot\theta\frac{\partial}{\partial\phi}\right)\left[-\sqrt{\frac{3}{8\pi}}\sin\theta e^{i\phi}\right] \\ &= \hbar e^{-i\phi}\sqrt{\frac{3}{8\pi}}\left[\cos\theta e^{i\phi} - i\cos\theta(i)e^{i\phi}\right] \\ &= \hbar\sqrt{\frac{3}{2\pi}}\cos\theta \end{aligned}$$

Thus

$$Y_1^0(\theta,\phi) = \sqrt{\frac{3}{4\pi}}\cos\theta$$

Then again for Y_1^{-1}

$$\begin{aligned} L_- Y_1^0(\theta,\phi) = \hbar\sqrt{2}Y_1^{-1}(\theta,\phi) &= -\hbar e^{-i\phi}\left(\frac{\partial}{\partial\theta} - i\cot\theta\frac{\partial}{\partial\phi}\right)\left[\sqrt{\frac{3}{4\pi}}\cos\theta\right] \\ &= -\hbar e^{-i\phi}\sqrt{\frac{3}{4\pi}}\left[-\sin\theta\right] \end{aligned}$$

And finally

$$Y_1^{-1} = \hbar\sqrt{\frac{3}{8\pi}}\sin\theta e^{-i\phi}$$

which obeys the fact that $Y_l^{-m} = (-1)^m(Y_l^m)^*$ and all of these agree with the listed Y_l^m functions in equation (12.5.39)

Exercise 12.5.12 Since L^2 and L_z commute with Π, they should share a basis with it. Verify that under parity $Y_l^m \to (-1)^l Y_l^m$. First show that $\theta \to \pi - \theta$, $\phi \to \phi + \pi$ under parity. Prove the result for Y_l^l. Verify that L_- does not alter parity thereby proving the result for all Y_l^m.) (Π is the parity operator)

Consider the spherical coordinates (where ϕ is the azimuthal angle and θ is the polar angle.

$$\begin{aligned} x &= r\cos\phi\sin\theta \\ y &= r\sin\phi\sin\theta \\ z &= r\cos\theta \end{aligned}$$

The parity operator exchanges a vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \to -\vec{r}$

Which makes our new spherical coordinates (r', θ', ϕ') have to obey the following restrictions

$$\begin{aligned} -\cos\phi'\sin\theta' &= \cos\phi\sin\theta \\ -\sin\phi'\sin\theta' &= \sin\phi\sin\theta \\ -\cos\theta' &= \cos\theta \end{aligned}$$

r' obviously equals r under parity. $-\cos\theta' = \cos\theta$ enforces $\theta' = \pi \pm \theta$. We remember that $\theta \in [0, \pi]$ so we discard the positive solution at $\theta' = \pi - \theta$.

Then $\sin(\theta') = \sin(\pi - \theta) = \sin\theta$ then $-\sin\phi' = \sin\phi$ allowing us to find that $\phi' = \phi + \pi$.

Now we prove the result for Y_l^l.

$$Y_l^l(\theta, \phi) = (-1)^l \left[\frac{(2l+1)!}{4\pi}\right]^{1/2} \frac{1}{2^l l!} \sin^l\theta e^{il\phi}$$

And under parity

$$\begin{aligned} Y_l^l(-\theta, -\phi) &= (-1)^l \left[\frac{(2l+1)!}{4\pi}\right]^{1/2} \frac{1}{2^l l!} \sin^l(\pi - \theta) e^{il(\phi+\pi)} \\ &= (-1)^l \left[\frac{(2l+1)!}{4\pi}\right]^{1/2} \frac{1}{2^l l!} (\sin\theta)^l e^{i\phi} (-1)^l \\ &= (-1)^l Y_l^l \end{aligned}$$

Of course, L_- in the coordinate basis does not care for the phase $(-1)^l$ so this makes the relation true for all Y_l^m

Exercise 12.5.13 Consider a particle in a state described by

$$\psi = N(x + y + 2z)e^{-\alpha r}$$

where N is a normalization factor.

Part 1 Show, by rewriting the $Y_1^{\pm 1,0}$ functions in terms of x, y, z and r, that

$$\begin{aligned} Y_1^{\pm 1} &= \mp\sqrt{\frac{3}{4\pi}} \frac{x \pm iy}{2^{1/2} r} \\ Y_1^0 &= \sqrt{\frac{3}{4\pi}} \frac{z}{r} \end{aligned}$$

We start from,

$$\begin{aligned} Y_1^{\pm 1} &= \mp\sqrt{\frac{3}{8\pi}} \sin\theta e^{\pm i\phi} \\ Y_1^0 &= \sqrt{\frac{3}{4\pi}} \cos\theta \end{aligned}$$

These coordinate transformations are clear from the substitutions

$$\begin{aligned} \sin\theta e^{\pm i\phi} &= \sin\theta\cos\phi + i\sin\phi\sin\theta = \frac{1}{r}(x + iy) \\ \cos\theta &= (r\cos\theta)/r = z/r \end{aligned}$$

Part 2 Using this result, show that for a particle described by ψ above, $P(l_z = 0) = 2/3$; $P(l_z = \pm\hbar) = 1/6$.

The question hints that we can express ψ as a linear combination of the spherical harmonics. This enforces that the coefficients squared of each eigenfunction gives us probabilities of the respective eigenvalue.

Indeed observe that

$$Y_1^1 + Y_1^{-1} = \sqrt{\frac{3}{4\pi}}\frac{\sqrt{2}}{r}x$$
$$Y_1^1 - Y_1^{-1} = \sqrt{\frac{3}{4\pi}}\frac{\sqrt{2}}{r}iy$$

Thus we can rewrite ψ

$$\begin{aligned}\psi &= Ne^{-\alpha r}\Big(x + y + 2z\Big) \\ &= Ne^{-\alpha r}r\sqrt{\frac{4\pi}{3}}\left((Y_1^1 + Y_1^{-1})\frac{1}{\sqrt{2}} + (Y_1^1 - Y_1^{-1})\frac{1}{i\sqrt{2}} + 2Y_l^0\right)\end{aligned}$$

Now the probabilities of a given eigenvalue is typically $P(l_z = l) = \frac{|\langle l|\psi\rangle|^2}{\langle\psi|\psi\rangle}$

But, we can thankfully ignore the normalization constant as for this exercise, as it suffices to find the relative probabilities and then normalize them afterwards.

$$P_{rel}(l_z = \pm\hbar) = |(1/\sqrt{2})(1 - i)|^2 = 1$$
$$P_{rel}(l_z = 0) = |2|^2 = 4$$

Normalizing the sum of probabilities to unity we get $P(l_z = \pm\hbar) = 1/6$ and $P(l_z = \pm\hbar) = 2/3$

Exercise 12.5.14 Consider the rotation $\theta_x\hat{i}$. Under this

$$\begin{aligned}x &\to x \\ y &\to y\cos\theta_x - z\sin\theta_x \\ z &\to z\cos\theta_x + y\sin\theta_x\end{aligned}$$

Therefore we must have

$$\psi(x, y, z) \xrightarrow[U(R(\theta_x\hat{i}))]{} \psi_r = \psi(x, y\cos\theta_x + z\sin\theta_x, z\cos\theta_x - y\sin\theta_x)$$

Let us verify this prediction for a special case

$$\psi = Aze^{-r^2/a^2}$$

which must go into $\psi_r = A(z\cos\theta_x - y\sin\theta_x)e^{-r^2/a^2}$

Part 1 Expand ψ in terms of $Y_1^{\pm 1,0}$.

Using the rewriting of the eignefunctions from the previous question, it is clear that ψ only has a component with Y_1^0

$$\psi = Ae^{-r^2/a^2} r\sqrt{\frac{4\pi}{3}} Y_1^0$$

Part 2 Use the matrix $e^{-i\theta_x L_x/\hbar}$ to find the fate of ψ under this rotation. Check your result against that anticipated above. [Hint: (1) ψ Y_1^0, which corresponds to [0, 1, 0]]

The rotation of ψ is $\psi_r = e^{-i\theta_x L_x/\hbar}\psi$.

First, lets find the operator L_x in the L_z eigenbasis.

$$\begin{aligned} L_x |1,1\rangle &= \frac{L_+ + L_-}{2} |1,1\rangle = \frac{\hbar}{\sqrt{2}} |1,0\rangle \\ L_x |1,0\rangle &= \frac{L_+ + L_-}{2} |1,0\rangle = \frac{\hbar}{\sqrt{2}} (|1,1\rangle + |1,-1\rangle) \\ L_x |1,-1\rangle &= \frac{L_+ + L_-}{2} |1,-1\rangle = \frac{\hbar}{\sqrt{2}} |1,0\rangle \end{aligned}$$

Hence the matrix representation of L_x in this basis is

$$L_x = \frac{\hbar}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

We use an expression from exercise 12.5.5

$$e^{-i\theta_x L_x^{(1)}/\hbar} = (\cos\theta_x - 1)\left(\frac{L_x^{(1)}}{\hbar}\right)^2 - i\sin\theta_x \left(\frac{L_x^{(1)}}{\hbar}\right) + \mathbb{1}^{(1)}$$

The above expression can be directly computable using $e^{\Omega} = \sum_{n=0}^{\infty} (n!)^{-1} \Omega^n$

We evaluate the rotation to be the desired expression.

$$\begin{aligned} \psi_r &= e^{-i\theta_x L_x/\hbar}\psi \\ &= \left[(\cos\theta_x - 1)\left(\frac{L_x^{(1)}}{\hbar}\right)^2 - i\sin\theta_x \left(\frac{L_x^{(1)}}{\hbar}\right) + \mathbb{1}^{(1)}\right]\left(Ae^{-r^2/a^2} r\sqrt{\frac{4\pi}{3}} Y_1^0\right) \\ &= Are^{-r^2/a^2}\sqrt{\frac{4\pi}{3}}\left[(\cos\theta_x - 1)\frac{L_x}{\hbar\sqrt{2}}(Y_1^1 + Y_1^{-1}) - i\sin\theta_x \frac{1}{\sqrt{2}}(Y_1^1 + Y_1^{-1}) + Y_1^0\right] \\ &= Are^{-r^2/a^2}\sqrt{\frac{4\pi}{3}}\left[(\cos\theta_x - 1)Y_1^0 - i\sin\theta_x \frac{1}{\sqrt{2}}(Y_1^1 + Y_1^{-1}) + Y_1^0\right] \\ &= Are^{-r^2/a^2}\sqrt{\frac{4\pi}{3}}\left[\cos\theta_x Y_1^0 - i\sin\theta_x \frac{1}{\sqrt{2}}(Y_1^1 + Y_1^{-1})\right] \\ &= A\left[z\cos\theta_x - y\sin\theta_x\right]e^{-r^2/a^2} \end{aligned}$$

Exercise 12.6.1 A particle is described by the wave function

$$\psi_E(r,\theta,\phi) = Ae^{-r/a_0}$$

Part 1 What is the angular momentum of the state?

$l = 0$ and $m = 0$ since ψ only contains radial components. Thus $R_{Elm} = \psi_E(r,\theta,\phi)$

Part 2 Assuming ψ_E is an eigenstate in a potential that vanishes as $r \to \infty$, find E. (Match leading terms in Schrodinger's Equation.)

Using the above in the radial equation,

$$\begin{aligned}
\left[-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{\partial}{\partial r}r^2\frac{\partial}{\partial r}\right]Ae^{-r/a_0} &= (E-V)Ae^{-r/a_0} \\
\left[-\frac{\hbar^2}{2\mu}\left(\frac{-2}{a_0 r}+\frac{1}{a_0^2}\right)\right]Ae^{-r/a_0} &= (E-V)Ae^{-r/a_0} \\
-\frac{\hbar^2}{2\mu}\left(\frac{-2}{a_0 r}+\frac{1}{a_0^2}\right) &= (E-V)
\end{aligned}$$

As $r \to \infty$, where $V(r) \to 0$ we see that $E = -\hbar^2/2\mu a_0^2$

Part 3 Having found E, consider finite r and find $V(r)$

Back to the radial equation we readily see that $V = -\hbar^2/\mu a_0 r$

Exercise 12.6.2 Provide the steps that derive

$$\left(\frac{d^2}{dr^2}+\frac{2\mu}{\hbar^2}\left[E-V(r)-\frac{l(l+1)}{2\mu r^2}\right]\right)U_{El} = 0$$

from (Eqn 12.6.3)

We start with (12.6.3) and substitute $R_{El} = U_{El}/r$

$$\begin{aligned}
\left(-\frac{\hbar^2}{2\mu}\left[\frac{1}{r^2}\frac{\partial}{\partial r}r^2\frac{\partial}{\partial r}-\frac{l(l+1)}{r^2}\right]+V(r)\right)R_{El} &= ER_{El} \\
-\frac{\hbar^2}{2\mu}\left[\frac{1}{r^2}\frac{\partial}{\partial r}(rU'_{El}-U_{El})-\frac{l(l+1)}{r^2}\frac{U_{El}}{r}\right]+V(r)\frac{U_{El}}{r} &= EU_{El}/r \\
-\frac{\hbar^2}{2\mu}\left[\frac{U''_{El}}{r}-\frac{l(l+1)}{r^2}\frac{U_{El}}{r}+V(r)\frac{U_{El}}{r}\right] &= EU_{El}/r \\
\left(\frac{d^2}{dr^2}+\frac{2\mu}{\hbar^2}\left[E-V(r)-\frac{l(l+1)}{2\mu r^2}\right]\right)U_{El} &= 0
\end{aligned}$$

Exercise 12.6.3 Show that Eq. (12.6.7b) follows from Eq. (12.6.7a).

This problem is a typical integration by parts problem where we enforce an equality by insisting the boundary conditions are 0. We start from the hermiticity of D_1

$$\int_0^\infty U_1^*(D_1U_2)dr = \int_0^\infty (D_1U_1)^*U_2 dr$$

We expand with the definition of D_1 and integrate by parts the hand side.

$$
\begin{aligned}
&= \int_0^\infty U_1^* \left(\left[-\frac{\hbar^2}{2\mu}\frac{d^2}{dr^2} + V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] U_2 \right) dr \\
&= -\frac{\hbar^2}{2\mu} \int_0^\infty U_1^* U_2'' dr + \int_0^\infty U_1^* \left[V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] U_2 dr \\
&= -\frac{\hbar^2}{2\mu} \left(U_1^* U_2' \Big|_0^\infty - \int_0^\infty U_1^{*'} U_2' \right) + \int_0^\infty U_1^* \left[V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] U_2 dr \\
&= -\frac{\hbar^2}{2\mu} \left(U_1^* U_2' \Big|_0^\infty - U_1^{*'} U_2 \Big|_0^\infty + \int_0^\infty U_1^{*''} U_2 dr \right) + \int_0^\infty U_1^* \left[V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] U_2 dr \\
&= -\frac{\hbar^2}{2\mu} \left(U_1^* U_2' - U_1^{*'} U_2 \right) \Big|_0^\infty + \int_0^\infty (D_1 U_1)^* U_2 dr
\end{aligned}
$$

Hence, the equality holds on the condition that the boundary conditions equal to zero.

Exercise 12.6.4 Show that

$$\delta^3(\vec{r} - \vec{r'}) = \delta(x - x')\delta(y - y')\delta(z - z') = \frac{1}{r^2 \sin\theta}\delta(r - r')\delta(\theta - \theta')\delta(\phi - \phi')$$

We have the definition of the Dirac delta function that $\int \delta(x - x')f(x')dx' = f(x)$

Thus if

$$
\begin{aligned}
\int \delta^3(\vec{r} - \vec{r'}) f(\vec{r'}) d^3\vec{r'} &= f(\vec{r}) \\
\int dx' \int dy' \int dz' \delta^3(\vec{r} - \vec{r'}) f(x', y', z') &= f(x, y, z)
\end{aligned}
$$

Then letting $\delta^3(\vec{r} - \vec{r'}) = \delta(x - x')\delta(y - y')\delta(z - z')$ is a sufficient condition by applying the definition across three separate integrals.

Similarly, doing the same thing in spherical coordiantes of $\vec{r}$

$$
\begin{aligned}
\int \delta^3(\vec{r} - \vec{r'}) f(\vec{r'}) d^3\vec{r'} &= f(\vec{r}) \\
\int d\theta' \int d\phi' \int dr' r'^2 \sin\theta' \delta^3(\vec{r} - \vec{r'}) f(r', \theta', \phi') &= f(r, \theta, \phi)
\end{aligned}
$$

Doing the same process, if we naively assumed $\delta^3(\vec{r} - \vec{r'}) = \delta(r - r')\delta(\theta - \theta')\delta(\phi - \phi')$ then the integral would evaluate to $r^2 \sin\theta f(r, \theta, \phi)$. It stands to reason that the Jacobian $r^2 \sin\theta$ needs to be divided out to get the expression we need.

Part 2 Show that

$$\nabla^2(1/r) = -4\pi\delta^3(\vec{r})$$

.

We first show that $\nabla^2(1/r) = 0$ if $r \neq 0$. We do this by applying the Laplacian ∇^2 in spherical coordinates

$$\nabla^2(1/r) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)\frac{1}{r} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(-1\right) = 0$$

To find out what happens at $r = 0$, we will consider a small sphere V with surface S centered at the origin and use Gauss' law (Divergence theorem) and the identity $\nabla^2\phi = \nabla \cdot \nabla\phi$

$$\begin{aligned}\int_V \nabla^2(1/r)d^3r = \int_V \nabla\cdot\nabla(1/r)d^3r &= \iint_S \nabla(1/r)\cdot d\vec{A} \\ &= -\iint_S \hat{r}/r \cdot d\vec{A} = -4\pi r^2/r^2 = -4\pi\end{aligned}$$

If $\nabla^2(1/r)$ integrates over all space to -4π but evaluates to zero everywhere else, then it must be equal to the dirac delta function centered at the origin times -4π.

Exercise 12.6.5 Show that D_l is nondegenerate in the space of functions U that vanish as $r \to 0$. (Recall the proof of Theorem 15, section 5.6) Notice that U_{El} is nondegenerate even for $E > 0$. This means that E, l, and m, label the state fully in three dimensions.

Th proof follows exactly form Theorem 15.

We assume that U_1 and U_2 are both eigenfunctions of D_l that vanish as $r \to 0$ and have eigenvalues E.

$$\begin{aligned}\left[-\frac{\hbar^2}{2\mu}\frac{d^2}{dr^2} + V(r) + \frac{l(l+1)}{2\mu r^2}\right]U_1 &= EU_1 \\ \left[-\frac{\hbar^2}{2\mu}\frac{d^2}{dr^2} + V(r) + \frac{l(l+1)}{2\mu r^2}\right]U_2 &= EU_2\end{aligned}$$

I multiply both equations by the other eigenfunction and subtract the two and get,

$$\begin{aligned}0 &= U_1''U_2 - U_1U_2'' \\ &= \frac{d}{dr}\left(U_1'U_2 - U_1U_2'\right) \\ c &= \left(U_1'U_2 - U_1U_2'\right)\end{aligned}$$

If $U_{1,2} \to 0$ as $r \to 0$ then $c = 0$. Then we integrate

$$\begin{aligned}dU_1/U_1 &= dU_2/U_2 \\ \ln|U_1| &= \ln|U_2| + d \\ U_1 &= e^d U_2\end{aligned}$$

Since the two eigen functions only different by a global phase, then they are functionally the same eigenfunction, thus proving the claim of nondegeneracy of D_l.

Exercise 12.6.6 Verify Eqs (12.6.21/22) are equivalent to Eq. (12.6.20)

The equations correspond to the raising and lowering operators,

$$d_l = \frac{d}{d\rho} + \frac{l+1}{\rho}$$
$$d_l^\dagger = -\frac{d}{d\rho} + \frac{l+1}{\rho}$$

Let us test $d_l d_l^\dagger U$

$$\begin{aligned} U = d_l d_l^\dagger U &= \left(\frac{d}{d\rho} + \frac{l+1}{\rho}\right)\left(-\frac{d}{d\rho} + \frac{l+1}{\rho}\right)U \\ &= -\frac{d^2}{d\rho^2}U + (l+1)\frac{d}{d\rho}\frac{U}{\rho} - \frac{l+1}{\rho}\frac{dU}{d\rho} + \frac{(l+1)^2}{\rho^2} \\ &= \left[-\frac{d^2}{d\rho^2} - \frac{l+1}{\rho^2} + \frac{(l+1)^2}{\rho^2}\right]U \\ &= \left[-\frac{d^2}{d\rho^2} + \frac{l(l+1)}{\rho^2}\right]U \end{aligned}$$

Which is exactly Eq. (12.6.20)

Part 2 Verify Eq. (12.6.24)

We write the equation with test function U, on the LHS

$$\begin{aligned} d_l^\dagger d_l U &= \left(-\frac{d}{d\rho} + \frac{l+1}{\rho}\right)\left(\frac{d}{d\rho} + \frac{l+1}{\rho}\right)U \\ &= -U'' - (l+1)\left(\frac{U}{\rho}\right)' + (l+1)\left(\frac{U'}{\rho}\right) + (l+1)^2\left(\frac{U}{\rho^2}\right) \end{aligned}$$

on the RHS

$$\begin{aligned} d_{l+1} d_{l+1}^\dagger U &= \left(\frac{d}{d\rho} + \frac{l+2}{\rho}\right)\left(-\frac{d}{d\rho} + \frac{l+2}{\rho}\right)U \\ &= -U'' + (l+2)\left(\frac{U}{\rho}\right)' - (l+2)\left(\frac{U'}{\rho}\right) + (l+2)^2\left(\frac{U}{\rho^2}\right) \end{aligned}$$

Combining both the sides verifies the equation.

$$\begin{aligned} 0 &= (2l+3)\left(\frac{U}{\rho}\right)' - (2l-3)\left(\frac{U'}{\rho}\right) + (2l+3)\left(\frac{U}{\rho^2}\right) \\ &= (2l+3)\frac{\rho U' - U}{\rho^2} - (2l-3)\left(\frac{U'}{\rho}\right) + (2l+3)\left(\frac{U}{\rho^2}\right) = 0 \end{aligned}$$

Exercise 12.6.7 Verify that j_0 and j_1 have limits given by

$$j_l \xrightarrow[\rho\to 0]{} \frac{\rho^l}{(2l+1)!!}$$

The first is immediately true via a common limit identity $j_0(\rho) = \frac{\sin\rho}{\rho} \xrightarrow[\rho\to 0]{} 1$

The second requires a Taylor expansion to the second order (first order terms cancel and go quickly to zero). Note that !! is the double factorial such that $z!! = z(z-2)(z-4)...$

$$\begin{aligned} j_1(\rho) &= \frac{\sin\rho}{\rho^2} - \frac{\cos\rho}{\rho} \\ &\approx \frac{\rho - \rho^3/3!}{\rho^2} - \frac{1+\rho^2/2}{\rho} = -\frac{\rho}{6} + \frac{\rho}{2} = \frac{\rho}{3} = \frac{\rho}{3!!} \end{aligned}$$

Exercise 12.6.8 Find the energy levels of a particle in a spherical box of radius r_0 in the $l = 0$ sector.

A spherical box implies a potential that is 0 when $r < r_0$ and is $V = \infty$ when $r > r_0$. This hints to us to make use of the boundary condition at $r = r_0$

Since $l = 0$, the angular momentum eigenfunction is constant and can just be absorbed in to the radial function's normalization, thus we get $\psi_{Elm} = R_{Elm}Y_l^m = NU_{El}/r$ and we progress using the radial equation

$$\left[\frac{d^2}{dr^2} + \frac{2\mu}{\hbar^2}\left(E - V(r) - \frac{l(l+1)\hbar^2}{2\mu r^2}\right)\right]U_{El} = 0$$

We take the case $V(r < r_0) = 0$ and plug in our $l = 0$ to get,

$$U''_{El} = -\frac{2\mu E}{\hbar^2}U_{El}$$

This has solutions of the form

$$U_{El} = Ae^{i\kappa x} + Be^{-i\kappa x}$$

where $\kappa = \sqrt{2\mu E/\hbar^2}$ and E is necessarily positive (larger than the minimium potential energy).

Our boundary condition $U_{El} \xrightarrow[r\to 0]{} 0$, enforced by the assumption that V(r) does not contain a delta function at the origin, posits that $A = -B$. Then the boundary condition $U_{El} \xrightarrow[r\to r_0]{} 0$, lets us get

$$U_{El} = N(e^{i\kappa r_0} - e^{-i\kappa r_0}) = 2iN\sin(\kappa r_0) = 0$$

This periodic condition allows us the find the allowed bound energy levels

$$r_0\sqrt{2\mu E/\hbar^2} = n\pi$$
$$E = \frac{n^2\pi^2\hbar^2}{2\mu r_0^2}$$

where $n = 1, 2...$, the n=0 solution is ignored as that would correspond to the trivial zero wavefunction solution.

Exercise 12.6.9 Show that the quantisation condition for $l = 0$ bound states in a spherical well of depth $-V_0$ and radius r_0 is

$$k'/\kappa = -\tan k'r_0$$

where k' is the wave number inside the well and $i\kappa$ is the complex wave number for the exponential tail outside. Show that there are no bound states for $V_0 < \pi^2\hbar^2/8\mu r^2$

Similar to the previous question, we set up the radial equation, absorbing the $l = 0$ angular eigenfunction into the radial equation, and solve depending on $V = -V_0$ when $r < r_0$ and $V = 0$ when $r > r_0$

The radial equation and its solution below is,

$$U''_{El} = -\frac{2\mu}{\hbar^2}(E - V(r))U$$

and,

$$U_{El} = Ae^{ikx} + Be^{-ikx}$$

where $k = \sqrt{2\mu(E - V(r)/\hbar^2}$

Inside the well $V = -V_0$, so our wavenumber becomes $k' = \sqrt{2\mu(E + V_0)/\hbar^2}$. Since E is necessarily $E > -V_0$ then our wavenumber is real and our exponential remain complex giving us harmonic solutions. We enforce like before $U_{El} \xrightarrow[r\to 0]{} 0$ to let $A = -B$ and we get the radial solution to inside the well,

$$U_{El(in)} = N_{in}(e^{ik'r} - e^{-ik'r}) = 2iN_{in}\sin(k'r)$$

We will also want to compute the derivative of this function as we need to ensure that the first derivative of the wave function is smooth across the boundary at $r = r_0$,

$$U'_{El(in)} = 2iN_{in}k'\cos(k'r)$$

Outside the well $V = 0$, so the wavenumber becomes $\kappa = \sqrt{2\mu E/\hbar^2}$. We only want to consider bound states which means we only want to look at cases where $E < 0$ which makes the solution's exponentials real. To ensure that U_{El} is normalizable, we also enforce boundary condition $U_{El} \xrightarrow[r\to\infty]{} 0$; immediately, we discard the coefficient of the exponential that blows up. Now our radial equation outside the well (for bound states) looks like

$$U_{El(out)} = N_{out} e^{-\kappa r}$$

And it's first derivative

$$U'_{El(out)} = -N_{out} \kappa e^{-\kappa r}$$

Finally equating the radial equations at $r = r_0$ gives us the relation,

$$2iN_{in} \sin(k' r_0) = N_{out} e^{-\kappa r_0}$$

And similarly, equating the first derivative equations to ensure smoothness gives the relation

$$2iN_{in} k' \cos(k' r_0) = -N_{out} \kappa e^{-\kappa r_0}$$

Combining these two relations gives us the desired quantisation condition

$$\frac{k'}{\kappa} = -\tan(k' r_0)$$

Exercise 12.6.11

Part 1 By combining Eq. (12.6.48) and (12.6.49), derive the two term recursion relation. Argue that $C_0 \neq 0$ if U is to have the right properties near $y = 0$. Derive the quantisations condition, Eq. (12.6.50)

We start with the isotropic oscillator's radial equation and the assumed form of auxiliary function $v(y)$

$$v'' - 2yv' + \left[2\lambda - 1 - \frac{l(l+1)}{y^2}\right] v = 0, \qquad \lambda = E/\hbar\omega$$

$$v(y) = \sum_{n=0}^{\infty} C_n y^{n+l+1}$$

We will let and combine the equations to get on the LHS

$$\sum_{n=0}^{\infty} C_n \Big((n+l)(n+l+1) y^{n+l-1} - 2(n+l+1) y^{n+l+1} + \big[2\lambda - 1\big] y^{n+l+1} - l(l+1) y^{n+l-1} \Big)$$

To get the recursion relation, we want to look at the coefficient of the y^{n+l+1} term and equate that to zero; we get

$$0 = C_{n+2}\big[(n+l+2)(n+l+3) - l(l+1)\big] + C_n\big[-2(n+l+1) + 2\lambda - 1\big]$$

or equivalently

$$\frac{C_{n+2}}{C_n} = \frac{3 + 2(n+l-\lambda)}{(n+l+2)(n+l+3) - l(l+1)}$$

Now, we can easily verify that the series will terminate at C_n if $3+2(n+l-\lambda) = 0$. We are now free to choose the energy (lambda) such that C_n is fixed as the terminus of the recursion relation.

Observe that if $C_0 = 0$ (and $C_1 \neq 0$ necessarily to avoid the trivial zero-wavefunction), we have $v(y) = C_1 y^{l+2} + C_3 y^{l+4} + ...$ Note that if $y \to 0$, then $v(y) \approx C_1 y^{l+2}$ thus $U \xrightarrow[y\to 0]{} C_1 e^{-y^2/2} y^{l+2} \approx y^{l+2}$. This does not obey the regular boundary condition of $U \sim y^{l+1}$. Hence we know that $C_0 \neq 0$.

We now set $C_1 = 0$ to avoid a non-terminating recursion relation for $v(y)$

this again has asymptotic nuance that I am skipping because I need to understand it further.

This means we can now only consider even n and the quantisation condition:

$$E = \hbar\omega(2n + l + 3/2) \quad n = 0, 1, 2...$$

Part 2 Calculate the degeneracy and parity at each n and compare with exercise 10.2.3, where the problem was solved in Cartesian coordinates.

Let us define the principal quantum number $N = 2n + l$. Since $l \geq 0$ we know that $l = N, N-2, N-4, ...$ and our energy formula is now $E = \hbar\omega(N + 3/2)$.

For a few examples let us illustrate, at $N = 0$, it is necessary that $l = 0$ and thus $m = 0$. At $N = 1$, it can be that $l = 1$ and $m = 0, \pm 1$. At $N = 2$, either $l = 0$ and $m = 0$ or $l = 2$ and $m = 0, \pm 1, \pm 2$

Thus, for a given principal quantum number N and thus a given energy, we have $2l+1$ states for every possible value of l. The degeneracy can be expressed with the sum

$$g_N = \sum_{l=N,N-2,...}^{0 \text{ or } 1} 2l + 1$$

The parity at each n can be considered how

We write the generic solution

$$\psi_{n,l,m} = \frac{1}{r} Y_l^m e^{-y^2/2} \sum_{n=0}^{2(\lambda-l)-3} C_{2n} y^{2n+l+1}$$

As we have shown in a previous exercise, in spherical coordinates the parity is demonstrated by negating the angular coordinates and not radial. Then we also showed that the parity of Y_l^m is $(-1)^l$. Since the angular coordinates is entirely contained in Y inside of ψ, then $\psi_{n,l,m}$ has parity $(-1)^l$ as well. Note that l has the same parity as N so the parity is equivalently $(-1)^N$.

Part 3 Construct the normalised eigenfunction ψ_{nlm} for $N = 0, 1$. Write them as linear combinations of the $N = 0$ and $n = 1$ eigenfunctions obtained in Cartesian coordinates.

For $N = 0$, $l = m = 0$. We can let the non coordinate factors get absorbed into the normalisation constant N. So

$$\psi_{000} = N \exp(-\frac{m\omega}{2\hbar} r^2)$$

We find N via the normalisation condition. The factor of a (1/2) comes from how (A.2.3) is defined across $\int_{-\infty}^{\infty}$, not $\int_0^{\infty}$

$$\begin{aligned} 1 = \int d^3r |N \exp(-m\omega r^2/2\hbar)|^2 &= |N|^2 \int d\phi \int d\theta \sin\theta \int_0^{\infty} r^2 \exp(-m\omega r^2/\hbar) dr \\ &= |N|^2 (4\pi) \frac{1}{2} \left(\frac{\hbar}{2m\omega}\right) \sqrt{\frac{\pi\hbar}{m\omega}} \end{aligned}$$

Hence,

$$\psi_{000} = \left(\frac{m\omega}{\pi\hbar}\right)^{3/4} e^{-m\omega r^2/2\hbar}$$

For $N = 1$, $l = 1$, and $m = 0, \pm 1$

So,

$$\psi_{110} = N \frac{1}{r} Y_1^m ...$$

And normalising again

$$\begin{aligned} 1 = \int d^3r |Nre^{-m\omega r^2/2\hbar}|^2 &= |N|^2 \int d\phi \int d\theta \sin\theta \int_0^{\infty} r^4 \exp(-m\omega r^2/\hbar) dr \\ &= |N|^2 (4\pi) \frac{1}{2} \frac{3}{4} \sqrt{\pi} \left(\frac{\hbar}{m\omega}\right)^{5/2} \end{aligned}$$

Thus

$$\psi_{100} = \sqrt{\frac{2}{3}} \pi^{-3/4} \left(\frac{m\omega}{\hbar}\right)^{5/4} re^{-m\omega r^2/2\hbar}$$